

Highlights

Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate

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- One fixed surrogate represents the complete recycling activated sludge system.
- Five controls and 110 mechanistic states are optimized in one sparse NLP.
- Split-conformal calibration bounds the admitted standardized state discrepancy.
- Smooth kinetics and Clarifier balances impose physics on the statistical response.
- One exact BDF replay validates the selected local candidate in every case.

Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate

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ABSTRACT

Activated sludge design and operation couple an inlet mixer, staged biological conversion, mixed-liquor recycle, return activated sludge, sludge wasting, and secondary clarification. Repeated dynamic solution of this closed mixer-reactor-Clarifier network is costly inside optimization. This study develops one simultaneous physics-constrained statistical nonlinear program for a prescribed N -stage system with a Clarifier; the case study fixes $N = 5$. Its 115 variables are five normalized design-and-operation controls and 110 reactor-and-Clarifier states. A fixed 351-feature, 170-response quadratic least-squares surrogate is trained only on a 14,000-state development block. An independent 2000-state iid block calibrates a finite-sample 95% conformal fidelity threshold, and an untouched 4000-state iid block gates raw predictive accuracy and operational coverage. During optimization, the complete 170-coordinate plant response is reconstructed from the 110 physical states and must satisfy both the calibrated surrogate-fidelity bound and the development leverage bound. The same variables satisfy all 110 scaled smooth ASM2d-TSN reactor and multilayer-Clarifier steady-state equations, physical bounds, two positive-solids guards, and five engineering inequalities. The objective is the engineering criterion alone. Nine deterministic CasADi-IPOPT starts provide one multistart local candidate per case, and one independent exact nonsmoothed BDF replay tests its residual, stability, domain, engineering, fidelity, and branch agreement. The method makes no global-optimality claim.

Nomenclature

Only nonstandard mathematical variables, sets, and operators are listed. Component names and process abbreviations are defined when introduced. All inequalities between vectors are componentwise. Whenever a numerical scaling expression compares a dimensional coordinate with a bare 1, the coordinate is first expressed in its tabulated unit; the 1 therefore means one of that stated unit. This convention creates dimensionless residuals without adding quantities of unlike dimensions.

$A \in \mathbb{R}^{K \times F}$	Full-row-rank stoichiometric-invariant operator
A_{cl}	Clarifier surface area
a_i	Dimensionless aeration setting in CSTR i
$\hat{C}$	Frozen standardized second-order coefficient matrix
$c_i \in \mathbb{R}_+^F$	Component concentration leaving CSTR i
$c_E, c_U \in \mathbb{R}_+^F$	Clarifier overflow and underflow concentrations
D_X, D_T	Positive development scales for states and effluent targets
D_f	Positive development scale for smooth mechanistic balances
D_θ	Diagonal design-and-operation decision-range matrix
$d(\theta, x, y), \delta$	Standardized surrogate-fidelity score and calibrated threshold
$\mathfrak{d}_i$	Recycle-adjusted dilution rate in CSTR i , d^{-1}
d_ϕ, L	Feature and complete-response dimensions
$\varepsilon_{mach} = 2^{-52}$	IEEE 754 binary64 machine epsilon
E_S, E_X	Selectors for soluble and particulate ASM components
F, K, n_θ, N, P	Numbers of components, invariants, decisions, CSTRs, and processes
$g_E = q_E c_E, g_U = q_U c_U$	Fresh-flow-normalized Clarifier outlet component mass flows
H	Total nominal hydraulic retention time based on fresh flow, h

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I_{comp}	Map from components to COD, TN, TP, and TSS
$J(\vartheta, \chi)$	Scalar engineering objective
$m \in \mathbb{R}_+^F$	Mixed component state entering CSTR 1
$\overline{M}_{\text{cl}} = M_{\text{cl}}/Q_0$	Fresh-flow-normalized Clarifier solids inventory
N_ℓ	Number of Clarifier layers
α_Q	Non-negative normalized effluent-quality weights
q_P, q_C, q_E, q_U	Flow ratios through the train, Clarifier, overflow, and underflow
Q_0	Fresh influent volumetric flow
r_I, r_R, w	MLR, RAS, and WAS flow ratios relative to Q_0
S_ϑ, S_x, S_ψ	Positive design, influent, and feature scales
μ_y, S_y	Mechanistic-state development mean and internal/branch scale
s_ℓ	TSS concentration in Clarifier layer ℓ
$t_X \in \mathbb{R}^F$	Column vector whose transpose maps components to TSS
T	Linear map selecting effluent-quality targets
$\tau_i = 24V_i/Q_0$	Fresh-flow HRT assigned to CSTR i , h
ϑ	Design-and-operation vector
$x \in \mathbb{R}_+^F$	Known fresh-influent component vector
z	Five normalized design-and-operation coordinates
$\chi, \hat{\chi}$	Reconstructed complete plant response and raw surrogate response
y	Mechanistic reactor-and-Clarifier dynamic state
η_j, v_j	Underflow recovery and densification factors for component j
$v \in \mathbb{R}^{P \times F}$	ASM2d-TSN stoichiometric matrix
$\phi(\vartheta, x)$	Dimensionless unique-monomial feature vector
ϱ_i	Mechanistic reaction-rate vector in CSTR i
ω_i	Net oxygen-transfer rate in CSTR i
$\mu_\vartheta, \mu_x, \mu_\psi, \mu_\chi$	Coordinate centers calculated from the applicable fitting rows
ϵ_{sm}	Dimensionless smooth-NLP regularization parameter
$\otimes$	Kronecker product

1. Introduction

Activated sludge plants are recycling systems. Mixed liquor may be returned to an upstream anoxic zone to make oxidized nitrogen available for denitrification, while settled biomass is returned from the Clarifier to maintain the biological inventory. A waste stream controls solids age, and the Clarifier overflow is the treated discharge. These streams are familiar individually, but changing one changes the hydraulic residence time, solids loading, and composition seen elsewhere in the loop.

Detailed activated sludge models represent these interactions through nonlinear reaction rates and clarification fluxes. They are indispensable for process analysis, yet repeated closed-loop steady-state solution can dominate the cost of optimization (de Araujo et al., 2013). Surrogate models can reduce that cost (Bhosekar and Ierapetritou, 2018), and component-space constraints can preserve mass conservation and non-negativity in activated-sludge regression (Selerio Jr., 2026). A complete recycling plant creates a broader risk: the optimizer can seek a small negative concentration, a broken recycle balance, or a favorable solids inventory that would never occur in the plant.

The recycle topology adds a second difficulty. If the reactors and Clarifier are predicted independently, their responses must be iterated around the MLR and RAS loops. A complete plant state is therefore represented by one system-level statistical response. During optimization that response is not accepted blindly: the 110 reactor-and-Clarifier states and all five controls are the variables of one constrained calculation, and the complete response is reconstructed from them. A unique-monomial quadratic basis and a single coefficient matrix make the statistical center identifiable, while a separately calibrated squared standardized-RMS fidelity threshold limits how far the reconstructed physical response may lie from that center.

The resulting contribution is a process-level simultaneous nonlinear-programming model. It compares feasible design-and-operation states using clarified effluent quality, reactor volume, aeration, pumping, and wasted solids; statistical fidelity and development support are feasibility requirements rather than objective penalties. The Clarifier

solids inventory is calculated from ten optimized layer concentrations. Smooth kinetics and Clarifier fluxes make the physical steady state differentiable inside the NLP, while one exact dynamic replay determines whether the selected point is reproduced by the original nonsmoothed model. The article defines the generator, estimator, conformal calibration, combined nonlinear program, multistart protocol, and final validation route as one self-contained calculation.

The mixer–reactor–Clarifier description covers a prescribed number N of biological stages; N is not an integer decision in the present optimization. The derivation applies to any fixed N , whereas the case study fixes five equal-volume CSTRs, two unaerated followed by three aerated. In the following sections, the equations are introduced in the order that an engineer can follow the water: fresh feed and recycles meet, mixed liquor passes through the reactors, solids and liquid divide in the Clarifier, and only then is the response scored.

2. Theory and Calculation

2.1. Walking Through the Recycling Process

Figure 1 is the physical picture behind every subsequent matrix. Fresh wastewater enters the mixer from the left. Both recycles return to that same point. The combined stream passes once through the N CSTRs. MLR is removed from the last-stage outlet before clarification; the remaining mixed liquor enters the Clarifier. Clarifier underflow is split into RAS and WAS without another composition-changing step, so RAS and WAS share the state c_U .

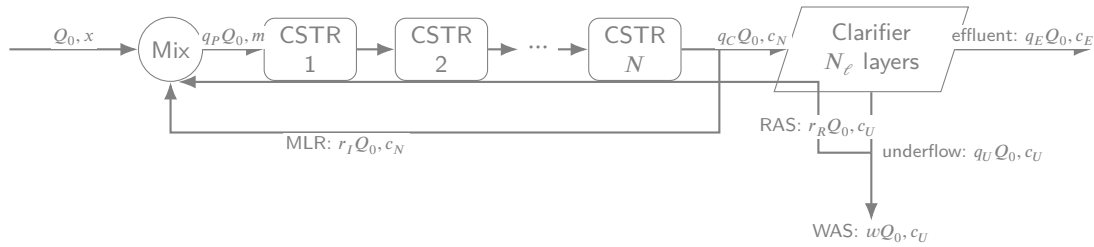


Figure 1: Stream-by-stream picture of the prescribed recycling N -stage activated sludge system. The MLR branch is taken before the Clarifier; RAS and WAS are hydraulic splits of one Clarifier underflow.

The fresh flow Q_0 is a convenient common ruler for the three manipulated side streams. Define

$$r_I = \frac{Q_I}{Q_0}, \quad r_R = \frac{Q_R}{Q_0}, \quad w = \frac{Q_W}{Q_0}, \quad (1)$$

where r_I is the MLR ratio, r_R is the RAS ratio, and w is the WAS ratio. The domain requires $r_I \geq 0$, $r_R > 0$, and $0 \leq w < 1$. These restrictions are not cosmetic: they keep every outlet flow and every concentration recovery division positive.

Starting at the mixer and following the arrows in Figure 1 gives four dimensionless throughflows:

$$\underbrace{q_P}_{\text{through CSTRs}} = 1 + r_I + r_R, \quad \underbrace{q_C}_{\text{into Clarifier}} = 1 + r_R, \quad \underbrace{q_U}_{\text{underflow}} = r_R + w, \quad \underbrace{q_E}_{\text{effluent}} = 1 - w. \quad (2)$$

The first identity counts fresh feed plus both returns. Removing MLR from the last-stage outlet leaves $q_C Q_0$. The Clarifier then divides that stream into $q_E Q_0$ and $q_U Q_0$, and the definitions show directly that $q_C = q_E + q_U$. This small hydraulic calculation is used repeatedly below; it is what makes RAS an internal stream and WAS an external loss.

Let m denote the mixer outlet, c_i the outlet of CSTR i , and c_U the Clarifier underflow. The mass entering the first tank must equal the fresh-feed mass plus the two returned masses, component by component:

$$q_P m = x + r_I c_N + r_R c_U. \quad (3)$$

Every term has the form “flow relative to Q_0 times concentration.” The factor Q_0 therefore cancels. Equation (3) is an identity, not a relationship to be learned. A topology with MLR withdrawn from another stage uses that stage state in place of c_N .

For a general prescribed topology, collect the stage volumes, aeration settings, and recycle decisions as

$$\vartheta = (\tau_1, \dots, \tau_N, a_1, \dots, a_N, r_I, r_R, w)^\top \in \mathcal{V}, \quad \tau_i = \frac{24V_i}{Q_0}. \quad (4)$$

Here τ_i is the nominal HRT in hours that tank i would provide at fresh flow alone. At fixed Q_0 , choosing τ_i chooses volume, $V_i = Q_0\tau_i/24$; it is therefore a design decision, not merely an operating knob. The liquid actually traverses the tank at $q_P Q_0$, so its one-pass residence time and dilution rate are

$$h_i = \frac{\tau_i}{q_P}, \quad \mathfrak{d}_i = \frac{24q_P}{\tau_i} \text{ d}^{-1}. \quad (5)$$

An engineer can read Equation (5) directly: increasing either recycle while tank volume is fixed moves more water through the same vessel and shortens one pass. The surrogate must see this coupling because the biology does.

2.2. From CSTR Balances to Inventory Rails

The next task is to state the physical equations that the optimization must satisfy. For CSTR i , steady accumulation is zero, so the difference between outlet and inlet transport must be supplied by reactions and oxygen transfer:

$$\mathfrak{d}_i(c_i - c_{i-1}) = v^\top \rho_i + \omega_i e_{S_O}, \quad c_0 = m. \quad (6)$$

The F entries of $c_i - c_{i-1}$ are component changes across the tank. The P entries of ρ_i are the ASM2d-TSN process rates, and the columns of $v^\top$ translate those rates into component production or consumption. The last term adds oxygen through aeration; e_{S_O} selects only the dissolved-oxygen coordinate.

The combined NLP carries a smooth form of this full rate vector, but invariant combinations remain valuable structural checks on the equations and reconstructed streams. Appendix A.2 constructs five physical conservation rows in B and normalizes each row separately to form $A \in \mathbb{R}^{K \times F}$. Separate normalization keeps the inert-soluble, nitrogen, phosphorus, alkalinity, and metal inventories as recognizable rails. Its required checks are

$$Av^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K, \quad \|A_{k,:}\|_2 = 1. \quad (7)$$

The first equality says that each row of A annihilates every reaction source; the second prevents external oxygen transfer from masquerading as a conserved inventory; full row rank removes duplication, and the final relation fixes each rail's scale without mixing it with another inventory.

Premultiplying Equation (6) by A removes both source terms. Because $\mathfrak{d}_i > 0$, the remaining statement is simply

$$Ac_1 = Am, \quad Ac_i = Ac_{i-1}, \quad i = 2, \dots, N. \quad (8)$$

These equalities are useful to picture as K rails extending through the train: individual COD, nitrogen, phosphorus, and biomass components may move between forms, but the selected stoichiometric inventories cannot jump from one tank to the next. They need not equal Ax , because RAS changes the material circulating inside the plant. The fresh-feed relation appears only after the external Clarifier and WAS boundaries are closed.

2.3. Following Components Through the Clarifier

The Clarifier changes where material leaves, not how much of a non-reactive component exists. Its component balance is therefore

$$q_C c_N = q_E c_E + q_U c_U. \quad (9)$$

The small underflow ratio can make c_U numerically large even when its component mass is ordinary. It is more stable and more intuitive to predict the outlet masses normalized by fresh flow:

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (10)$$

Equation (9) then becomes

$$g_E + g_U = q_C c_N. \quad (11)$$

Once q_E and q_U are known, the outlet concentrations are recovered uniquely as $c_E = g_E/q_E$ and $c_U = g_U/q_U$. Thus, non-negative mass-flow coordinates also give non-negative outlet concentrations.

Clarification must not be represented as solids generation. The TSS map satisfies $t_X \in \mathbb{R}_+^F$, $E_S t_X = 0$, and $\|t_X\|_1 > 0$. Once the development block D described below has been generated, define the frozen reporting scale

$$S_{f,j} = \max \left(1, \max_{i \in D} |q_{C,i} c_{N,ij}| \right) \quad (12)$$

in the tabulated mass-flow-coordinate unit. A particulate recovery is reported only when $q_C c_{N,j} > 10^{-10} S_{f,j}$. For such a coordinate define the underflow recovery and concentration ratio by

$$\eta_j = \frac{g_{U,j}}{q_C c_{N,j}}, \quad v_j = \frac{c_{U,j}}{c_{N,j}} = \eta_j \frac{q_C}{q_U}. \quad (13)$$

The first factor, η_j , is the fraction of incoming particulate mass sent downward. Non-negativity and Clarifier balance give $0 \leq \eta_j \leq 1$; the densification rule sharpens this to $q_U/q_C \leq \eta_j \leq 1$ and therefore $1 \leq v_j \leq q_C/q_U$. The factor q_C/q_U expresses how recovered mass is concentrated into a smaller water stream. At or below the declared threshold, neither ratio is identifiable and it is reported as unavailable; the scale and availability flag are persisted.

The soluble and particulate coordinates require different physical rules. Let E_S select the ten soluble components and E_X the ten particulate components. Ideal soluble pass-through means that the underflow carries the same soluble concentration as the feed:

$$E_S g_U = q_U E_S c_N. \quad (14)$$

Together with Equation (11), this also gives $E_S c_E = E_S c_U = E_S c_N$. For particulates, the mechanistic Clarifier moves a common floc solids flux. A successfully reconstructed and validated state must exhibit the physically necessary direction of that movement component by component:

$$E_X g_U \geq q_U E_X c_N. \quad (15)$$

Equivalently, every particulate underflow concentration is at least its feed concentration. Because $g_E \geq 0$ and Equation (11) holds, recovery cannot exceed the incoming particulate mass. In the combined formulation, these relationships follow from the layer-based component reconstruction when its physical profile is reproduced; they are replayed as diagnostics rather than introduced as substitute kinetic constraints.

A single learned Clarifier inventory would leave no way to verify where the solids reside. The response therefore contains the N_ℓ layer TSS concentrations

$$s = (s_1, \dots, s_{N_\ell})^\top, \quad (16)$$

ordered from the top water surface to the bottom hopper. The two end layers are tied to the outlet TSS values,

$$q_E s_1 = t_X^\top g_E, \quad q_U s_{N_\ell} = t_X^\top g_U, \quad (17)$$

and an accepted exact state is checked against the physically interpretable top-to-bottom envelope,

$$s_1 \leq s_\ell \leq s_{N_\ell}, \quad \ell = 2, \dots, N_\ell - 1. \quad (18)$$

The Clarifier solids inventory is now a consequence of a pictured profile rather than an independent output:

$$M_{cl} = \sum_{\ell=1}^{N_\ell} V_{cl,\ell} s_\ell, \quad \overline{M}_{cl} = \frac{M_{cl}}{Q_0}. \quad (19)$$

The stream bookkeeping can now be closed around the entire plant. Premultiply the mixer balance by A , use the reactor rails in Equation (8), substitute the Clarifier balance, and cancel MLR and RAS because both remain inside the boundary. What remains is

$$Ax = q_E A c_E + w A c_U = A g_E + \frac{w}{q_U} A g_U. \quad (20)$$

This is the engineer's external check: conserved inventory enters only with fresh wastewater and leaves only with treated overflow and wasted sludge.

2.4. One Identifiable Response for the Complete Loop

Separate unit surrogates would have to be iterated around two recycle paths. Instead, collect every quantity needed to describe the closed loop in one response:

$$\chi = \begin{bmatrix} m \\ c_1 \\ \vdots \\ c_N \\ g_E \\ g_U \\ s \end{bmatrix} \in \mathbb{R}^L, \quad L = (N + 3)F + N_\ell. \quad (21)$$

For the five-CSTR, 20-component, ten-layer case, $L = 170$. Concentrations, normalized mass flows, and layer TSS have different units, so the regression is built in standardized coordinates. Let

$$\tilde{\vartheta} = S_\vartheta^{-1}(\vartheta - \mu_\vartheta), \quad \tilde{x} = S_x^{-1}(x - \mu_x), \quad \tilde{\chi} = D_\chi^{-1}(\chi - \mu_\chi), \quad (22)$$

where all centers and positive diagonal scales come only from the development rows.

A second-order model needs squares and cross-products, but it needs each monomial only once. Let $q(v)$ list $v_j v_k$ for $j \leq k$, and form the nonconstant vector

$$\psi(\vartheta, x) = \begin{bmatrix} \tilde{\vartheta} \\ \tilde{x} \\ q(\tilde{\vartheta}) \\ q(\tilde{x}) \\ \tilde{\vartheta} \otimes \tilde{x} \end{bmatrix}. \quad (23)$$

Centering and scaling these nonconstant terms once more gives the serialized feature vector

$$\phi(\vartheta, x) = \begin{bmatrix} 1 \\ S_\psi^{-1}[\psi(\vartheta, x) - \mu_\psi] \end{bmatrix}, \quad d_\phi = 1 + n_\vartheta + F + \frac{n_\vartheta(n_\vartheta + 1)}{2} + \frac{F(F + 1)}{2} + n_\vartheta F. \quad (24)$$

For $n_\vartheta = 5$ and $F = 20$, the basis contains $d_\phi = 351$ columns. It includes every linear term, square, within-vector cross-product, and design–influent interaction exactly once. Within $q(v)$ the pairs are serialized lexicographically as $(1, 1), (1, 2), \dots, (1, d), (2, 2), \dots, (d, d)$; the Kronecker block is serialized with the influent index varying fastest.

The frozen raw prediction is

$$\tilde{\chi}_{\text{raw}} = \hat{C}\phi(\vartheta, x), \quad \hat{\chi}(\vartheta, x) = \mu_\chi + D_\chi \tilde{\chi}_{\text{raw}}. \quad (25)$$

The matrix $\hat{C} \in \mathbb{R}^{L \times d_\phi}$ is the complete input–response map. Its design matrix must pass the full-column-rank and condition-number tests stated in the training procedure. For a fixed influent, Equation (25) is a vector-valued quadratic in the design and operating decisions.

2.5. Reconstructing the Complete Response from Physical States

The surrogate supplies a statistical center, but it does not supply the state variables used by the optimizer. Those variables are the NF reactor concentrations and N_ℓ Clarifier-layer concentrations,

$$y = (c_1^\top, \dots, c_N^\top, s_1, \dots, s_{N_\ell})^\top \in \mathbb{R}_+^{NF+N_\ell}. \quad (26)$$

For the case study, $y \in \mathbb{R}_+^{110}$. Given (ϑ, y, x) , take c_N directly from the final reactor block and set $s_F = t_x^\top c_N$. Soluble components pass through the Clarifier, while every particulate component follows the common layer-TSS proportions:

$$\begin{aligned} c_{E,j} &= c_{U,j} = c_{N,j}, & j &\in S, \\ c_{E,j} &= \frac{s_1}{s_F} c_{N,j}, & c_{U,j} &= \frac{s_{N_\ell}}{s_F} c_{N,j}, & j &\in \mathcal{X}. \end{aligned} \quad (27)$$

The optimization guard $s_F \geq 1$ makes every displayed ratio well defined at a feasible point; a finite C^2 reciprocal extension used during infeasible trial evaluations is specified below. The remaining response blocks follow without another solve:

$$g_E = q_E c_E, \quad g_U = q_U c_U, \quad m = \frac{x + r_I c_N + r_R c_U}{q_P}, \quad \chi = \chi(\vartheta, y; x) = \begin{bmatrix} m \\ c_1 \\ \vdots \\ c_N \\ g_E \\ g_U \\ s \end{bmatrix}. \quad (28)$$

This construction is the bridge between the 110-state physics and the 170-response statistics. It closes the mixer and Clarifier component balances, preserves soluble pass-through, and ties both outlet TSS concentrations and the Clarifier inventory to the layer profile. The reactor and layer balances introduced below then determine whether those reconstructed streams form a steady physical plant.

The statistical model is used as a calibrated feasibility envelope around this reconstructed response. Define

$$d(\vartheta, x, y) = \frac{1}{L} \left\| D_\chi^{-1} [\chi(\vartheta, y; x) - \hat{\chi}(\vartheta, x)] \right\|_2^2. \quad (29)$$

Here D_χ contains only development-set target scales. The calibration procedure later supplies a finite, strictly positive threshold δ . Requiring $d \leq \delta$ keeps the physical state within the calibrated standardized neighborhood of the statistical center. It is a constraint, not an objective term. Split-conformal coverage applies to exchangeable future design rows and is not claimed for an optimizer-selected input.

2.6. From a Physical State to an Engineering Decision

The discharge is c_E , not the mixed liquor c_N . Let T select the effluent components or composites to be scored, let D_T contain frozen positive development-response scales, and let $\alpha_Q \geq 0$ with $\mathbf{1}^\top \alpha_Q = 1$. A dimensionless concentration-quality term is

$$J_Q(\chi) = \alpha_Q^\top D_T^{-1} T c_E = \alpha_Q^\top D_T^{-1} T \frac{g_E}{q_E}. \quad (30)$$

Concentration is used because the primary case represents discharge-quality limits; the corresponding loads $Q_0 T g_E$ are retained and reported separately. Let $J_{\text{des/op}}$ contain the declared volume, aeration, pumping, and wasted-solids contributions. The complete engineering score is

$$J(\vartheta, \chi) = \lambda_Q J_Q(\chi) + J_{\text{des/op}}(\vartheta, \chi), \quad \lambda_Q > 0. \quad (31)$$

Equipment and permit limits answer whether a simultaneous state may be operated. Denote the five engineering inequalities by $g_{\text{eng}}(\vartheta, \chi) \leq 0$. Whole-plant SRT is one such quantity. Because the Clarifier inventory is derived from the layer profile, it is

$$\text{SRT}(\vartheta, \chi) = \frac{\sum_{i=1}^N V_i t_X^\top c_i + \sum_{\ell=1}^{N_\ell} V_{\text{cl},\ell} s_\ell}{Q_0 (q_E t_X^\top c_E + w t_X^\top c_U)}. \quad (32)$$

The numerator is all suspended solids held in the CSTRs and Clarifier. The denominator contains only solids that cross the plant boundary in effluent and WAS. MLR and RAS are absent because they circulate internally. In the NLP, positive-solids domain guards are imposed before SRT is used, and the 8–30 d limits are written by cross multiplication. Consequently, the numerical formulation never divides by a trial-state boundary-solids loss.

Let $D_\vartheta = \text{Diag}(\vartheta^U - \vartheta^L)$ and normalize the five controls by

$$z = D_\vartheta^{-1} (\vartheta - \vartheta^L) \in [0, 1]^5. \quad (33)$$

The normalized controls and the internally scaled physical states are the only optimization variables. The statistical response is never allowed to move independently of a mechanistic state: it is always the reconstruction $\chi[\vartheta(z), y; x]$. The full combined NLP is stated after its smooth mechanistic equations and numerical scales have been defined.

2.7. Local-Solution Interpretation

The raw surrogate is quadratic in z , while the flow-dependent balances contain products of controls and state coordinates. Kinetic rates, SRT restrictions, and Clarifier fluxes are nonlinear. The simultaneous problem is therefore nonconvex. Nine deterministic starts are solved independently, and only the best accepted local result is reported. No start, first-order certificate, or comparison among nine starts establishes global optimality.

3. Methodology

3.1. End-to-End Study Design

The computational study follows four information-separated phases. First, the coupled ASM2d-TSN–Clarifier model generates one fixed 20,000-state design. Second, three independently generated blocks provide 14,000 development rows, 2000 calibration rows, and 4000 assessment rows. The estimator, preprocessing, state scales, objective scales, and smooth-equation scales are fitted once using development rows only; calibration fixes δ , and the untouched assessment rows gate the frozen raw response. Third, one combined physics-constrained statistical NLP is solved from nine deterministic starts for every nominal, robustness, and sensitivity case. Fourth, its one selected local candidate receives one exact nonsmoothed dynamic replay. Calibration, assessment, and optimization responses never refit the estimator or alter any scale, feature, weight, tolerance, or case definition.

3.2. Coupled Reactor–Clarifier Data Generator

The reaction model is ASM2d with two-step nitrification (ASM2d-TSN), comprising $P = 28$ processes and $F = 20$ components (Guerrero et al., 2011; Henze et al., 2006). Appendix A gives every rate, stoichiometric coefficient, composition coefficient, and parameter value used here. Every matrix, prediction, and physical check uses the fixed order

$$(S_O, S_F, S_A, S_{NH_4}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{PO_4}, S_I, S_{ALK}, \\ X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH}). \quad (34)$$

The first ten entries are soluble and the last ten particulate. Component units remain those of the mechanistic model; COD, TN, TP, and TSS are calculated only after a complete component state is available through the fixed I_{comp} matrix.

The case study fixes $N = 5$ equal-volume CSTRs. CSTRs 1–2 are unaerated denitrification stages, CSTRs 3–5 share one manipulated aerobic intensity, and MLR is taken from CSTR 5. Appendix A.2 defines a deterministic five-row physical conservation matrix B and its row-normalized form A . For a matrix $C \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, define $\text{rank}_\epsilon(C)$ as the number satisfying $\sigma_k > \max(m, n)\epsilon_{\text{mach}}\sigma_1$. Generation stops unless $\text{rank}_\epsilon(v) = 15$, $\text{rank}_\epsilon(B) = \text{rank}_\epsilon(A) = 5$, and $\|Av^\top\|_\infty$, $\|Ae_{S_O}\|_\infty$, and $\max_k \| \|A_{k,:}\|_2 - 1 \|$ are each no larger than 10^{-10} .

The last CSTR feeds a non-reactive, one-dimensional $N_\ell = 10$ layer Clarifier based on the flux-limited clarification–thickening model of Takács et al. (1991) and the benchmark conventions of Alex et al. (2018) and Jeppsson et al. (2007). To make the numerical generator reproducible, let s_ℓ be layer TSS, $s_F = t_X^\top c_5$ the feed TSS, and

$$v_s(s) = \max\{0, \min[v_0^{\max}, v_0(e^{-r_h(s-s_{\min})} - e^{-r_p(s-s_{\min})})]\}, \quad s_{\min} = f_{ns}s_F. \quad (35)$$

The two exponentials describe hindered settling at high solids and the low-concentration branch; the outer bounds prevent a negative settling speed or a speed above $v_0^{\max}$. Define the unrestricted gravity flux $j_s(s) = v_s(s)s$. At the interface between top-to-bottom layers ℓ and $\ell + 1$, the benchmark receiver-capacity limiter is

$$j_{\ell+1/2}^{\text{set}} = \begin{cases} j_s(s_\ell), & s_{\ell+1} \leq X_t, \\ \min\{j_s(s_\ell), j_s(s_{\ell+1})\}, & s_{\ell+1} > X_t. \end{cases} \quad (36)$$

This expression lets dilute receiving layers accept the incoming gravity flux and limits that flux when the receiving layer is already concentrated.

Take downward flux as positive and let the feed enter layer $\ell_F = 5$ in a top-to-bottom, one-based array (the fifth layer from the top and sixth from the bottom). Above the feed, bulk liquid moves upward at $v_E = Q_0 q_E / A_{\text{cl}}$; below it, bulk liquid moves downward at $v_U = Q_0 q_U / A_{\text{cl}}$. The total internal-interface TSS flux is therefore

$$\mathcal{J}_{\ell+1/2} = \begin{cases} -v_E s_{\ell+1} + j_{\ell+1/2}^{\text{set}}, & \ell < \ell_F, \\ v_U s_\ell + j_{\ell+1/2}^{\text{set}}, & \ell \geq \ell_F. \end{cases} \quad (37)$$

Table 1

Fixed multilayer-Clarifier geometry and settling parameters.

Quantity	Value	Unit or convention
Surface area A_{cl}	1500	m ²
Total depth	4.0	m
Layer count; depth; volume	10; 0.4; 600	–; m; m ³
Feed layer	5 from top	6 from bottom; array index 4
Maximum settling speed $v_0^{\max}$	250	m d ^{–1}
Theoretical coefficient v_0	474	m d ^{–1}
Hindered coefficient r_h	0.000576	m ³ g ^{–1}
Low-concentration coefficient r_p	0.00286	m ³ g ^{–1}
Non-settleable fraction f_{ns}	0.00228	dimensionless
Flux-limiting threshold X_l	3000	g TSS m ^{–3}
Blanket diagnostic threshold	3000	g TSS m ^{–3}

At the water surface and hopper, $J_{1/2} = -v_E s_1$ and $J_{N_\ell+1/2} = v_U s_{N_\ell}$; no gravity flux is allowed to cross those physical boundaries. The finite-volume layer balance is

$$V_{cl,\ell} \frac{ds_\ell}{dt} = A_{cl}(J_{\ell-1/2} - J_{\ell+1/2}) + \mathbf{1}_{\{\ell=\ell_F\}} Q_0 q_C s_F. \quad (38)$$

Summing Equation (38) over all layers cancels internal fluxes and recovers $q_C s_F = q_E s_1 + q_U s_{N_\ell}$ at steady state. This is why the endpoint equations in the simultaneous physical constraints are not arbitrary additions.

Soluble components follow the liquid phase unchanged. Particulate components share the TSS flux and retain the Clarifier-feed particulate proportions. The complete layer reconstruction is

$$c_{cl,\ell,j} = \begin{cases} c_{5,j}, & j \in S, \\ (c_{5,j}/s_F)s_\ell, & j \in \mathcal{X}, s_F > 0, \\ 0, & j \in \mathcal{X}, s_F = 0, \end{cases} \quad J_{\ell+1/2,j} = (c_{5,j}/s_F)J_{\ell+1/2} \quad (j \in \mathcal{X}, s_F > 0). \quad (39)$$

Thus $c_{E,j} = (c_{5,j}/s_F)s_1$ and $c_{U,j} = (c_{5,j}/s_F)s_{N_\ell}$ for particulates. Because every particulate entry of t_X is positive, $s_F = 0$ in a non-negative state implies that all particulate feed coordinates are zero, so the limiting definition is mass consistent.

The mechanistic dynamic state is $y = (c_1, \dots, c_5, s_1, \dots, s_{10}) \in \mathbb{R}_+^{110}$. At every residual evaluation, Equation (39) gives c_E and c_U , and Equation (3) gives m . The transient reactor balance whose steady form is Equation (6) is

$$\frac{dc_i}{dt} = \mathbf{d}_i(c_{i-1} - c_i) + v^\top \rho(c_i) + \omega_i e_{S_O}, \quad c_0 = m. \quad (40)$$

The five vector derivatives in Equation (40) and the ten scalar layer derivatives in Equation (38) therefore form one coupled function $\dot{y} = \mathcal{F}_{\text{dyn}}(y; \vartheta, x)$. Eliminating the mixer and outlet algebraic variables in this way does not break the loop: a change in the bottom layer immediately changes RAS, the mixer, CSTR 1, and every downstream state in the same residual evaluation.

The nonlinear residual is expressed in dimensionless-state rates. Reactor component (i, j) is divided by $c_j^{\text{ref}} = \max(1, x_j^U)$, where x_j^U is the upper bound in Table 4; every layer derivative uses one fixed start-specific reference $s_0^{\text{ref}} = \max[1, t_X^\top c_5(y_0)]$. These values form the positive diagonal dynamic scale S_{dyn} . The scaled derivative replay used for steady-state acceptance evaluates the same component references and the current Clarifier-feed TSS reference. This scaling gives one tolerance the same engineering meaning for dissimilar quantities such as dissolved oxygen and sludge-blanket solids.

Two initial states are prescribed. Start 1 uses $c_i = x$ for every stage and $s_\ell = t_X^\top x$ for every layer. Start 2 uses

$$c_i = E_S^\top E_S x + \alpha_i E_X^\top E_X x, \quad (\alpha_1, \dots, \alpha_5) = (1.5, 2.0, 2.5, 3.0, 3.5), \quad (41)$$

and, after forming $s_F = t_X^\top c_5$,

$$(s_1, \dots, s_{10}) = s_F (0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00). \quad (42)$$

Table 2

Mechanistic steady-state numerical and physical acceptance contract.

Check or setting	Limit	Purpose
Nonlinear xtol , ftol , gtol	10^{-9}	Steady nonlinear solve
Maximum residual evaluations	5000 per start	Bounded retry effort
BDF relative; scaled absolute tolerance	10^{-7} ; 10^{-9}	Dynamic relaxation
BDF horizon; maximum step	$T_{\text{rel}} = \max(400, 50/w) \text{ d}$; $T_{\text{rel}}/100$	WAS-scaled relaxation limit
Scaled CSTR/layer derivative	$\leq 10^{-8} \text{ d}^{-1}$	Local balance closure
Scaled early-stop derivative	$\leq 10^{-9} \text{ d}^{-1}$	Early terminal event before full replay
Minimum state coordinate	$\geq -10^{-10}$	No clipped negative state
Largest real reduced-Jacobian eigenvalue	$\leq 10^{-8} \text{ d}^{-1}$	Local stability screen
Component Clarifier and plant-boundary residual	$\leq 10^{-8}$ scaled	External closure

Each prescribed initial state y_0 is first relaxed in scaled coordinates by integrating

$$\frac{d\tilde{y}}{dt} = S_{\text{dyn}}^{-1} \mathcal{F}_{\text{dyn}}(S_{\text{dyn}}\tilde{y}; \vartheta, x), \quad \tilde{y}(0) = S_{\text{dyn}}^{-1} y_0, \quad (43)$$

with variable-order BDF. The relaxation horizon follows the slow solids-removal mode,

$$T_{\text{rel}} = \max(400, 50/w) \text{ d}, \quad (44)$$

and the adaptive maximum step is $T_{\text{rel}}/100$. The factor $50/w$ gives the slow solids mode enough time to satisfy both the local derivative and accumulated plant-boundary closure tests as wasting approaches its lower bound; it is an upper limit, not a forced simulation duration. Integration terminates early when the scaled derivative infinity norm reaches 10^{-9} d^{-1} . Relative and scaled-coordinate absolute tolerances are 10^{-7} and 10^{-9} .

A successful integrator status is not by itself a positivity certificate. Every stored physical state on the scaled-state trajectory is checked and must be strictly positive. If an internal evaluation fails or any stored state is non-positive, the same start is retried with

$$y^{\log} = \log \tilde{y}, \quad \frac{dy^{\log}}{dt} = \text{Diag}(e^{y^{\log}})^{-1} S_{\text{dyn}}^{-1} \mathcal{F}_{\text{dyn}}(S_{\text{dyn}} e^{y^{\log}}; \vartheta, x), \quad (45)$$

which keeps every evaluated physical state on that route positive without clipping. The relaxed endpoint is replayed against the complete 10^{-8} acceptance contract below. Only an endpoint that misses that contract is supplied to a non-negative bounded trust-region-reflective least-squares calculation as a local polish, using $\text{xtol}=\text{ftol}=\text{gtol}=10^{-9}$ and at most 5000 residual evaluations. Start 1 fixes the branch when this complete route is accepted; Start 2 is used only after the complete Start-1 route fails. No preceding Latin-hypercube solution initializes another row.

For the stability screen, the reduced Jacobian is $J_y = \partial \mathcal{F}_{\text{dyn}} / \partial y$ after the algebraic quantities have been eliminated. Column j is calculated by a centered finite difference with step $\sqrt{\varepsilon_{\text{mach}}} \max(1, |y_j|)$, using a forward difference only when the centered step would cross zero. The prescribed initialization route defines the mechanistic branch represented in the data; neither simultaneous NLP is assumed to make that nonlinear branch unique.

For every algebraic balance written as $\sum_k t_k = 0$, its scaled replay residual is $|\sum_k t_k| / \max(1, \sum_k |t_k|)$. The 10^{-8} external-closure limit in Table 2 applies to this quantity. Every accepted state also passes hydraulic splits, RAS/WAS common composition, soluble pass-through, $q_U/q_C - 10^{-10} \leq \eta_j \leq 1 + 10^{-10}$ for every available particulate ratio, the layer envelope, and finite-rate checks. The lower recovery bound is the mechanistic counterpart of $c_{U,j} \geq c_{5,j}$ and makes an accepted target feasible for the simultaneous-state inequalities to the stated tolerance. Every ρ_p , oxygen-transfer term, settling velocity, and interface flux must be finite and no smaller than -10^{-12} when its mathematical definition is non-negative. A violation invalidates the state; negative quantities are not clipped into acceptance.

3.3. Case-Study Design, Operating, and Influent Domains

The primary case reduces the general vector in Equation (4) to five decisions:

$$\vartheta = (H, a, r_I, r_R, w)^\top. \quad (46)$$

The total nominal HRT is divided equally, so $\tau_i = H/5$ and $V_i = Q_0 H / (24 \times 5)$. Aeration is zero in CSTRs 1–2 and $k_L a = 47a \text{ d}^{-1}$ in CSTRs 3–5. Table 3 is both the training and optimization box; no surrogate response outside it is scored.

Table 3

Fixed topology, joint design-and-operation domain, and engineering screens.

Quantity	Value or range	Interpretation
Fresh flow Q_0	10,000 m ³ d ⁻¹	Plant throughput basis
Number of CSTRs N	5	Fixed topology
Nominal total HRT H	[6, 36] h	Sets five equal tank volumes
Aerobic setting a	[0, 1]	$k_I a = 47a$ d ⁻¹ in CSTRs 3–5
MLR ratio r_I	[0, 4]	Q_I/Q_0
RAS ratio r_R	[0.25, 1.25]	Q_R/Q_0
WAS ratio w	[0.001, 0.05]	Q_W/Q_0
Dissolved-oxygen saturation	8.5 g O ₂ m ⁻³	Aeration boundary
SRT acceptance interval	[8, 30] d	Engineering capacity constraint
Clarifier surface overflow	≤ 20 m d ⁻¹	Engineering hydraulic screen
Clarifier feed solids loading	≤ 100 kg TSS m ⁻² d ⁻¹	Engineering solids screen
Underflow TSS $X_U^{\max}$	≤ 15,000 g TSS m ⁻³	Pumpability screen

Table 4

Fresh-influent component ranges for mechanistic design and surrogate fitting.

Component	Range	Unit	Component	Range	Unit
S_O	[0.0, 0.5]	g O ₂ m ⁻³	S_I	[10, 90]	g COD m ⁻³
S_F	[20, 180]	g COD m ⁻³	S_{ALK}	[1.6, 5.2]	mol HCO ₃ ⁻ m ⁻³
S_A	[5, 80]	g COD m ⁻³	X_I	[20, 120]	g COD m ⁻³
S_{NH4}	[12, 55]	g N m ⁻³	X_S	[60, 280]	g COD m ⁻³
S_{NO2}	[0, 3]	g N m ⁻³	X_H	[15, 100]	g COD m ⁻³
S_{NO3}	[0, 8]	g N m ⁻³	X_{PAO}	[5, 60]	g COD m ⁻³
S_{N2}	[0, 2]	g N m ⁻³	X_{PP}	[2, 20]	g P m ⁻³
S_{PO4}	[2, 18]	g P m ⁻³	X_{PHA}	[1, 30]	g COD m ⁻³
			X_{AOB}	[0.5, 8]	g COD m ⁻³
			X_{NOB}	[0.5, 8]	g COD m ⁻³
			X_{MeP}	[0, 12]	g TSS m ⁻³
			X_{MeOH}	[0, 12]	g TSS m ⁻³

The surface-overflow and solids-loading screens are

$$\text{SOR} = \frac{Q_0 q_E}{A_{cl}}, \quad \text{SLR} = \frac{Q_0 q_C t_X^T c_5}{1000 A_{cl}}. \quad (47)$$

The factor 1000 converts grams to kilograms in SLR. These are case-study engineering screens (Tchobanoglous et al., 2014), not fitted Takács parameters. At fixed Q_0 and area, SOR cannot become active over the declared WAS range but is retained as a hydraulic safety check. The primary underflow-TSS limit is 15,000 g TSS m⁻³; the precisely matched 12,000 and 20,000 g TSS m⁻³ sensitivity cases are defined with the objective-weight cases below.

Table 4 completes the 25-dimensional training domain. The alkalinity range is expressed in the ASM unit mol HCO₃⁻ m⁻³: 80–260 g CaCO₃ m⁻³ converts to 1.6–5.2 mol of charge m⁻³, which is the range used in the component vector.

The nominal influent is the componentwise midpoint of Table 4, in the order of Equation (34):

$$x_{\text{nom}} = (0.25, 100, 42.5, 33.5, 1.5, 4, 1, 10, 50, 3.4, 70, 170, 57.5, 32.5, 11, 15.5, 4.25, 4.25, 6, 6)^T. \quad (48)$$

It is a designed in-domain influent, not a measured plant sample. Over the declared box,

$$q_P \in [1.25, 6.25], \quad H/5 \in [1.2, 7.2] \text{ h}, \quad H/(5q_P) \in [0.192, 5.76] \text{ h}. \quad (49)$$

These are derived hydraulic supports of the joint design rather than additional acceptance constraints.

3.4. Mechanistic Design and Stored Targets

The stored design concatenates three independently generated blocks in the fixed 25-dimensional order ($H, a, r_I, r_R, w, x_1, \dots$) a 14,000-point strength-1 Latin-hypercube development block (McKay et al., 1979), a 2000-point iid-uniform calibration block, and a 4000-point iid-uniform assessment block. Their separate SplitMix64 scalar states start at

42, 43, and 44, respectively. The random streams are defined here so that no software-specific seeding convention is required. All generator variables are unsigned 64-bit integers; $\oplus$ is bitwise exclusive-or, $\gg$ is a logical right shift, and every addition and multiplication is reduced modulo 2^{64} . From the applicable block state q_0 , draw the word $\mathcal{U}_t$ by

$$q_{t+1} = q_t + 9E3779B97F4A7C15_{16}, \quad (50)$$

$$v_1 = [q_{t+1} \oplus (q_{t+1} \gg 30)] BF58476D1CE4E5B9_{16}, \quad (51)$$

$$v_2 = [v_1 \oplus (v_1 \gg 27)] 94D049BB133111EB_{16}, \quad (52)$$

$$\mathcal{U}_t = v_2 \oplus (v_2 \gg 31). \quad (53)$$

The draw counter advances once for every generated word, including a rejected shuffle word. For development dimension j , initialize $\pi_j = (0, \dots, n_D - 1)$ with $n_D = 14,000$. For the separate shuffle index $k = n_D - 1, \dots, 1$, let $b = k + 1$ and $L_b = 2^{64} - (2^{64} \bmod b)$; draw until $\mathcal{U}_t < L_b$ and swap entries k and $\mathcal{U}_t \bmod b$. Here L_b is an exact mathematical integer, permitted to equal 2^{64} and therefore exempt from the unsigned-64-bit storage statement; generator states and generated words remain unsigned 64-bit integers. This preserves the stated rejection sequence and is unbiased because L_b is divisible by b . After that Fisher–Yates permutation, draw once for each development row in increasing order and set $\xi_{ij} = (\mathcal{U}_t \gg 11)/2^{53}$. The development coordinate is

$$\zeta_{ij} = \frac{\pi_j(i-1) + \xi_{ij}}{n_D}, \quad i = 1, \dots, n_D. \quad (54)$$

For calibration and assessment, reset to the block's stated seed, traverse rows first and the displayed dimensions second, draw exactly one word per coordinate, and set

$$\zeta_{ij}^{\text{iid}} = \frac{(\mathcal{U}_t \gg 12) + 1/2}{2^{52}} \in (0, 1). \quad (55)$$

The 52-bit midpoints are exactly representable in binary⁶⁴, including the largest value $1 - 2^{-53}$, so no iid coordinate lies on a box face. Affine range maps send all unit coordinates to physical variables. Conditional on the development block, the calibration rows and a future iid-uniform row are exchangeable; the independent assessment block is untouched by fitting and calibration. Every point must produce an accepted state under Table 2. A failed point is not replaced. If any point remains unresolved after the fixed retry route, the study stops without a fitted response. All generator states, draw counts, start attempts, and diagnostics are retained. The robustness design uses its separate stream stated later.

For each accepted row, the target vector is

$$\chi_{\text{ASM}} = (m, c_1, \dots, c_5, g_E, g_U, s_1, \dots, s_{10}) \in \mathbb{R}^{170}. \quad (56)$$

The full component state of every Clarifier layer, c_E , c_U , component and TSS recoveries, composites, loadings, SRT, reaction and oxygen-transfer rates, initialization route, stability result, residuals, and solve time are retained as diagnostics. Outlet mass flows are modeled because their scale remains well behaved when q_U is small; the layer TSS values make M_{cl} a derived quantity through Equation (19).

The three blocks retain their immutable concatenated order. Rows 1–14,000 form $\mathcal{D}$, rows 14,001–16,000 form $\mathcal{C}$, and rows 16,001–20,000 form $\mathcal{A}$. This 70/10/20 membership is fixed before any response is examined. Only $\mathcal{D}$ may determine preprocessing, scales, coefficients, and numerical row scales. Set $\mathcal{C}$ is used only once to fix δ , and $\mathcal{A}$ remains untouched until the frozen estimator and threshold have been sealed. No row outside $\mathcal{D}$ enters a refit, and no candidate model is selected.

Two smaller profiles exercise the same block-generation, fitting, and combined-NLP contracts before the full study. The unit profile has 600 newly generated rows in three independent blocks, partitioned as 420 development, 60 calibration, and 120 assessment rows, with seeds 60042, 60043, and 60044; 420 exceeds the 351 feature columns. It evaluates 14 cases, attempts $9(14) = 126$ combined-NLP starts, and requires at most $600 + 14 = 614$ BDF routes. The `test_2000` profile has 1400, 200, and 400 rows with seeds 200042, 200043, and 200044, respectively, and runs ten robustness cases in addition to the nominal and 12 sensitivity cases. It therefore evaluates 23 cases, attempts $9(23) = 207$ combined-NLP starts, and requires at most $2000 + 23 = 2023$ BDF routes. Each profile regenerates its three blocks at its declared sizes and profile-specific seeds; it is not an ordered prefix or subsample of the full design. These profiles verify execution only and do not contribute numerical results to the full-study tables.

3.5. Fixed Development-Only Multiresponse Least-Squares Estimator

For the fixed fitting set $\mathcal{I} = \mathcal{D}$, containing $n = 14,000$ rows, define the population mean, population standard deviation, and numerical reference magnitude of a scalar coordinate v by

$$\bar{v}_{\mathcal{I}} = \frac{1}{n} \sum_{i \in \mathcal{I}} v_i, \quad s_{\mathcal{I}}(v) = \left[\frac{1}{n} \sum_{i \in \mathcal{I}} (v_i - \bar{v}_{\mathcal{I}})^2 \right]^{1/2}, \quad a_{\mathcal{I}}(v) = \max \left(1, \max_{i \in \mathcal{I}} |v_i| \right). \quad (57)$$

A coordinate is numerically variable only when

$$s_{\mathcal{I}}(v) > 10^{-12} a_{\mathcal{I}}(v). \quad (58)$$

This rule is applied to every design coordinate, influent coordinate, nonconstant monomial, and target coordinate. All 350 nonconstant feature coordinates and all 170 target coordinates must pass; the leading constant feature is retained by definition. Otherwise the fit is invalid.

The scaling matrices used throughout are fixed explicitly by the development rows:

$$\begin{aligned} S_{\vartheta} &= \text{Diag}\{s_{\mathcal{D}}(\vartheta_j)\}, & S_x &= \text{Diag}\{s_{\mathcal{D}}(x_j)\}, \\ S_{\psi} &= \text{Diag}\{s_{\mathcal{D}}(\psi_j)\}, & D_{\chi} &= \text{Diag}\{s_{\mathcal{D}}(\chi_j)\}. \end{aligned} \quad (59)$$

The corresponding $\mu_{\vartheta}, \mu_x, \mu_{\psi}, \mu_{\chi}$ are the development means in Equation (57), and $s_{\chi,j} := s_{\mathcal{D}}(\chi_j) = (D_{\chi})_{jj}$. No later block changes these matrices or centers.

Using only rows in $\mathcal{I}$, construct all centers and scales in Equations (22)–(24). Let $Y_{\mathcal{I}}$ contain the mechanistic targets from Equation (56), in the same row order as the features. The fitting matrices are

$$\Phi_{\mathcal{I}} = \begin{bmatrix} \phi_1^{\top} \\ \vdots \\ \phi_n^{\top} \end{bmatrix} \in \mathbb{R}^{n \times d_{\phi}}, \quad \tilde{Y}_{\mathcal{I}} = (Y_{\mathcal{I}} - \mathbf{1}_n \mu_{\chi}^{\top}) D_{\chi}^{-1} \in \mathbb{R}^{n \times L}, \quad (60)$$

where $d_{\phi} = 351$ and $L = 170$. The fixed estimator is

$$\hat{C}_{\mathcal{I}} = \arg \min_{C \in \mathbb{R}^{L \times d_{\phi}}} \frac{1}{nL} \left\| \tilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^{\top} \right\|_F^2. \quad (61)$$

Every output is estimated from the single fixed standardized design in Equation (60). There is no regularization choice, response-specific feature selection, hyperparameter search, or later refit.

A thin singular-value decomposition

$$\Phi_{\mathcal{I}} = U \Sigma V^{\top}, \quad \sigma_1 \geq \cdots \geq \sigma_{d_{\phi}} \geq 0, \quad (62)$$

defines

$$\tau_{\text{rank}} = \max(n, d_{\phi}) \varepsilon_{\text{mach}} \sigma_1, \quad \kappa_2(\Phi_{\mathcal{I}}) = \frac{\sigma_1}{\sigma_{d_{\phi}}}. \quad (63)$$

The design is admissible only when

$$\sigma_{d_{\phi}} > \tau_{\text{rank}}, \quad \kappa_2(\Phi_{\mathcal{I}}) \leq 10^8. \quad (64)$$

Under these conditions Equation (61) has one unique least-squares minimizer supported by the observed design.

The coefficients are calculated by column-pivoted Householder QR,

$$\Phi_{\mathcal{I}} \Pi = Q R_{\Phi}, \quad \hat{C}_{\mathcal{I}}^{\top} = W_{\text{QR}} = \Pi R_{\Phi}^{-1} Q^{\top} \tilde{Y}_{\mathcal{I}}. \quad (65)$$

The triangular solution is followed by one iterative-refinement replay in the same factorization. With $R_Y^{(0)} = \Phi_{\mathcal{I}} W_{\text{QR}} - \tilde{Y}_{\mathcal{I}}$, solve

$$\Delta W = \Pi R_{\Phi}^{-1} Q^{\top} (-R_Y^{(0)}), \quad W_{\text{QR}} \leftarrow W_{\text{QR}} + \Delta W. \quad (66)$$

This correction does not change the least-squares estimator; it suppresses triangular-solve roundoff before the normal-equation audit, which is especially important when a response is reproduced nearly exactly. The independent SVD calculation is

$$W_{\text{SVD}} = V \Sigma^{-1} U^T \tilde{Y}_I. \quad (67)$$

With $R_Y = \Phi_I W_{\text{QR}} - \tilde{Y}_I$, define

$$\eta_{\text{opt}} = \frac{\|\Phi_I^T R_Y\|_F}{\max\{1, \|\Phi_I\|_2 \|R_Y\|_F\}}, \quad \eta_{\text{agree}} = \frac{\|W_{\text{QR}} - W_{\text{SVD}}\|_F}{\max(1, \|W_{\text{SVD}}\|_F)}, \quad \max(\eta_{\text{opt}}, \eta_{\text{agree}}) \leq 100\kappa_2(\Phi_I)\epsilon_{\text{mach}}. \quad (68)$$

All coefficients must also be finite. Failure of Equations (58), (64), or (68) invalidates the response; no alternate estimator is substituted.

The development rows also determine the mechanistic residual and smoothing scales defined with the combined NLP below. They are calculated from named physical terms rather than from nearly zero fitted residuals, then frozen before calibration, assessment, or optimization is opened.

3.6. Computational Feasibility Gate

The full study has 113 cases: one nominal case, 100 fresh-influent robustness cases, two additional underflow-TSS cases, and ten objective-weight cases. Each case uses nine starts of the single combined NLP. Define the complete logical workloads

$$\begin{aligned} N_{\text{BDF}}^{\text{max}} &= 20,000 + 113 = 20,113, \\ N_{\text{combined-NLP}}^{\text{sci}} &= 9(113) = 1017. \end{aligned} \quad (69)$$

The first term in $N_{\text{BDF}}^{\text{max}}$ is data generation. The remaining term reserves one independent replay for the selected combined-NLP result in every case. This is the planned maximum: a case with no accepted NLP result has no decision to replay, so its realized count is smaller and is reported. An invocation counts once even when the prescribed internal Start-1/conditional-Start-2 generator route is needed.

The first 256 design rows are executed as an uncontended serial timing preflight while remaining part of the immutable design; their complete-route times define $t_{\text{BDF},95}$ by the nearest-rank rule. After calibration and assessment pass, a fixed 23-case panel preflights and caches every one of the nine combined-NLP starts. The panel contains the nominal case, all 12 nominal-influent sensitivity cases, and ten robustness influents selected from the full robustness design. To select those ten, standardize the 20 influent coordinates by their configured spans, initialize the comparison set with the nominal influent, and repeatedly add the unselected robustness row maximizing its minimum Euclidean distance to the nominal and already selected rows. An exact tie uses the smaller original robustness-row index; selection stops at ten or when all available rows have been used. Thus the reduced profile with ten robustness rows executes all 23 cases in the panel.

The nearest-rank 95th percentile over the panel's per-start durations defines $t_{\text{NLP},95}$. Panel results are scientific results and are reused rather than rerun. With the observed one-time fit duration t_{fit} , the empirical single-thread projection is

$$T_{\text{proj}} = t_{\text{fit}} + 1.5[20,113t_{\text{BDF},95} + 1017t_{\text{NLP},95}]. \quad (70)$$

If M_{fit} , M_{BDF} , and M_{NLP} are the maximum measured resident memories, set

$$M_{\text{proj}} = 1.5 \max(M_{\text{fit}}, M_{\text{BDF}}, M_{\text{NLP}}). \quad (71)$$

The factor 1.5 is a declared safety factor applied to both timed workloads and observed memory. Proceeding beyond each preflight requires $T_{\text{proj}} \leq 30$ core-days and $M_{\text{proj}} \leq 25$ GiB on the measured 31.43-GiB host. The run record stores the exact physical-memory byte count and host identity. These are empirical projections from the sampled design rows and case panel, not runtime or memory guarantees; heterogeneous cases outside the panel can take longer or use more memory. Failure stops the remaining study, while timing never changes the fit, blocks, starts, tolerances, or workload.

3.7. Conformal Calibration and Untouched Assessment

The frozen development response predicts every calibration row without physical optimization. For $i \in C$, define the nonconformity score exactly as

$$d_i = \frac{1}{L} \left\| D_{\chi}^{-1} [\chi_{ASM,i} - \hat{\chi}(\vartheta_i, x_i)] \right\|_2^2. \quad (72)$$

Let $d_{[1]} \leq \dots \leq d_{[n_C]}$ be the sorted scores. With $\alpha = 0.05$, the zero-based stored-array index and corresponding one-based order statistic are

$$k_{\delta} = \min\{n_C, \lceil (n_C + 1)(1 - \alpha) \rceil\}, \quad j_{\delta} = k_{\delta} - 1, \quad \delta = d_{[k_{\delta}]}. \quad (73)$$

Here k_{δ} is the one-based order-statistic index and j_{δ} is its zero-based stored-array index. For the full-study $n_C = 2000$, $k_{\delta} = 1901$, $j_{\delta} = 1900$, and $\delta = d_{[1901]}$; the same formula, without retuning, is used by a verification profile. Calibration fails unless every d_i and δ is finite and $0 < \delta \leq 1$. Conditional on the development fit and under the stated iid sampling model, before the pseudorandom calibration and future-row streams are fixed, split conformal gives finite-sample marginal coverage of at least $1901/2001 > 0.95$ for $d_{\text{new}} \leq \delta$ (Lei et al., 2018). This is not coverage conditional on the realized δ or on passing assessment, and it is not a probability guarantee for the resulting deterministic streams, an adaptively optimized z , or the simultaneously chosen 170-dimensional state. In optimization, δ is a predeclared squared standardized-RMS fidelity threshold, whose RMS radius is $\sqrt{\delta}$.

Only after $\hat{C}$, all development scales, and δ are sealed are the 4000 assessment rows opened. For assessment row i and coordinate j , let $e_{ij} = \hat{\chi}_{ij} - \chi_{ASM,ij}$. Raw-response metrics are

$$\text{RMSE}_j = \left[\frac{1}{n_A} \sum_{i \in A} e_{ij}^2 \right]^{1/2}, \quad \text{MAE}_j = \frac{1}{n_A} \sum_{i \in A} |e_{ij}|, \quad (74)$$

$$\text{Bias}_j = \frac{1}{n_A} \sum_{i \in A} e_{ij}, \quad \text{nRMSE}_j = \frac{\text{RMSE}_j}{s_{\chi,j}}, \quad (75)$$

$$R_j^2 = 1 - \frac{\sum_{i \in A} e_{ij}^2}{\sum_{i \in A} (\chi_{ASM,ij} - \bar{\chi}_{A,j})^2}. \quad (76)$$

For the R^2 denominator, Equation (58) is applied to target coordinate j with $\mathcal{I} = \mathcal{A}$; a failing denominator is reported as undefined. For a coordinate block B , the standardized block error is

$$E_B^{\text{raw}} = \left[\frac{1}{n_A |B|} \sum_{i \in A} \sum_{j \in B} \left(\frac{e_{ij}}{s_{\chi,j}} \right)^2 \right]^{1/2}. \quad (77)$$

It is reported for the mixer, every CSTR, both outlet-flow blocks, the layer profile, and the complete state. Overflow and underflow COD, TN, TP, and TSS and normalized Clarifier inventory use the same RMSE, MAE, bias, nRMSE, and R^2 definitions. Each derived-response standard deviation is calculated on $\mathcal{D}$ and must pass Equation (58) before it is used as an nRMSE denominator; failure invalidates the assessment.

Assessment fidelity coverage is

$$\hat{p}_A = \frac{1}{4000} \sum_{i \in A} \mathbf{1}_{\{d_i \leq \delta\}}, \quad (78)$$

where assessment d_i uses Equation (72) without recalibration. The frozen response is admitted to optimization only when every raw prediction is finite, $E_{\{1, \dots, L\}}^{\text{raw}} < 1$, and $\hat{p}_A \geq 0.90$. The coverage threshold is a predeclared operational gate, not a significance test: split conformal controls marginal coverage over calibration and future-row randomness, not a calibration-conditional Bernoulli probability fixed at 0.95. All other assessment metrics are descriptive. Failure stops the study. Assessment results never alter the estimator, δ , scales, objective, constraints, starts, or solver settings.

3.8. Nominal Objective and Constraints

For Equation (48), set $T = I_{\text{comp}}$ and $\alpha_Q = \frac{1}{4}\mathbf{1}_4$. Let D_T contain development-set standard deviations of overflow COD, TN, TP, and TSS. Each of these four derived coordinates must pass Equation (58); otherwise optimization does not begin. The primary engineering objective is

$$J = 0.50 \alpha_Q^\top D_T^{-1} I_{\text{comp}} c_E + 0.15 \frac{H - H^L}{H^U - H^L} + 0.20 a + 0.05 \frac{r_I - r_I^L}{r_I^U - r_I^L} + 0.05 \frac{r_R - r_R^L}{r_R^U - r_R^L} + 0.05 \frac{w t_X^\top c_U}{w^U(15,000)}. \quad (79)$$

The final term is proportional to the wasted-solids mass rate $Q_0 w t_X^\top c_U$; penalizing w alone would treat a dilute and a concentrated waste stream as equally costly. The constant Q_0 cancels in the normalization. All six coefficients are declared before optimization and sum to one. They define a transparent dimensionless case, not a claim of universal economic equivalence.

The weight sensitivity is designed to move one engineering preference at a time without inadvertently changing the total emphasis on effluent quality. Let

$$\beta = (\beta_H, \beta_a, \beta_I, \beta_R, \beta_W) = (0.15, 0.20, 0.05, 0.05, 0.05), \quad \beta_\Sigma = \sum_j \beta_j = 0.50. \quad (80)$$

For each $k \in \{H, a, I, R, W\}$ and $\gamma \in \{0.5, 1.5\}$, define one case by

$$\beta_k^{(k,\gamma)} = \gamma \beta_k, \quad \beta_j^{(k,\gamma)} = \beta_j \frac{\beta_\Sigma - \gamma \beta_k}{\beta_\Sigma - \beta_k} \quad (j \neq k). \quad (81)$$

Thus there are exactly ten one-at-a-time weight cases. The quality coefficient remains 0.50, the five non-quality coefficients remain positive and sum to 0.50, and every scale in Equation (79) remains fixed. Reporting the six unweighted normalized objective components alongside their coefficients makes the resulting tradeoff visible even though the scalar objectives of differently weighted cases are not directly comparable.

The underflow-TSS comparison contains exactly 12,000, 15,000, and 20,000 g TSS m⁻³. The 15,000 g TSS m⁻³ entry reuses the primary nominal case; only the underflow engineering constraint changes for the two additional entries, and the wasted-solids denominator remains 15,000 g TSS m⁻³ in all three. Each of these two cases and each of the ten cases in Equation (81) uses the nominal influent, frozen development estimator and scales, and the same nine-start combined-NLP protocol. Each selected result receives the same independent exact BDF validation as its corresponding nominal or robustness result.

The development design defines the leverage support screen. With Φ_D denoting the 14,000-row development feature matrix,

$$\ell(\vartheta, x) = \phi(\vartheta, x)^\top (\Phi_D^\top \Phi_D)^{-1} \phi(\vartheta, x), \quad \ell_{\max} = \max_{i \in D} \ell(\vartheta_i, x_i). \quad (82)$$

The displayed inverse defines the scalar; triangular solves with the frozen development QR factor are used numerically.

For a smooth physical domain, define $s_F = t_X^\top c_N$, the boundary solids concentration $B_X = q_E t_X^\top c_E + w t_X^\top c_U$, and total inventory $M_X = \sum_i V_i t_X^\top c_i + \sum_\ell V_{\text{cl},\ell} s_\ell$. Require the two separate domain guards

$$s_F \geq 1 \text{ g TSS m}^{-3}, \quad B_X \geq 1 \text{ g TSS m}^{-3}. \quad (83)$$

With the fixed development scale

$$S_M = \max \left(1, \max_{i \in D} \{ M_{X,i}, 30 Q_0 B_{X,i} \} \right), \quad (84)$$

the five engineering rows are

$$g_{\text{eng}} = \begin{bmatrix} (8Q_0 B_X - M_X)/S_M \\ (M_X - 30Q_0 B_X)/S_M \\ (\text{SOR} - 20)/20 \\ (\text{SLR} - 100)/100 \\ (t_X^\top c_U - X_U^{\max})/15000 \end{bmatrix} \leq 0. \quad (85)$$

The first two rows are the cross-multiplied 8–30 d SRT interval and never divide by a trial-state solids loss. The two domain guards are separate from these five engineering rows and are independently replayed; no accepted iterate or state is clipped. No additional discharge constraint is active; COD, TN, TP, and TSS remain in the objective.

The combined NLP has only two statistical restrictions: fidelity $d/\delta - 1 \leq 0$ and leverage $(\ell - \ell_{\max})/\max(\ell_{\max}, 10^{-12}) \leq 0$. Both are hard feasibility requirements. Neither is added to the engineering objective.

3.9. Smooth Mechanistic Equations for the Combined NLP

The five controls and 110 physical states in Equation (26) must satisfy all reactor and layer balances in the same optimization that enforces statistical fidelity. For stable numerical scaling, let μ_y and $s_{y,j}$ be the development means and population standard deviations of these coordinates, and define

$$S_y = \text{Diag}\{\max(1, s_{y,j})\}_{j=1}^{110}, \quad \tilde{y} = S_y^{-1}(y - \mu_y). \quad (86)$$

The internal primal is $(z, \tilde{y})$; every physical equation and reported quantity is evaluated after reconstructing $y = \mu_y + S_y \tilde{y}$. The mixer and all outlet states are reconstructed algebraically from (z, y, x) using the flow equations. The reactor block imposes every steady component balance in Equation (6); the ten layer rows impose Equation (38). Hence this formulation moves the reactor concentrations and Clarifier blanket together with the controls rather than repeatedly calling a dynamic simulator at trial controls.

The original Clarifier contains min–max branches that are unsuitable for a twice differentiable interior-point calculation. For quantities a and b sharing positive scale S , define

$$\begin{aligned} \text{smax}_{\epsilon, S}(a, b) &= \frac{1}{2} \left[a + b + \sqrt{(a - b)^2 + (\epsilon S)^2} \right], \\ \text{smmin}_{\epsilon, S}(a, b) &= \frac{1}{2} \left[a + b - \sqrt{(a - b)^2 + (\epsilon S)^2} \right], \end{aligned} \quad \epsilon = \epsilon_{\text{sm}} = 10^{-8}. \quad (87)$$

Each primitive differs from the corresponding exact maximum or minimum by no more than $\epsilon S/2$. Settling velocities use $S_v = v_0 = 474 \text{ m d}^{-1}$, and solids fluxes use $S_J = v_0(15,000) = 7.11 \times 10^6 \text{ g TSS m}^{-2} \text{ d}^{-1}$. To keep the exponentials bounded even at an infeasible trial layer below $s_{\min}$, define

$$S_\Delta = \max \left(1, \max_{\substack{i \in D \\ \ell = 1, \dots, 10}} |s_{i,\ell} - f_{ns} s_{F,i}| \right), \quad \Delta^{\text{sm}}(s) = \frac{1}{2} \left[s - s_{\min} + \sqrt{(s - s_{\min})^2 + (\epsilon S_\Delta)^2} \right]. \quad (88)$$

The smooth settling law is

$$v_s^{\text{sm}}(s) = \text{smax}_{\epsilon, S_v} \left(0, \text{smmin}_{\epsilon, S_v} \left[v_0^{\max}, v_0 \{ e^{-r_h \Delta^{\text{sm}}(s)} - e^{-r_p \Delta^{\text{sm}}(s)} \} \right] \right), \quad j_s^{\text{sm}}(s) = s v_s^{\text{sm}}(s). \quad (89)$$

Because $\Delta^{\text{sm}} \geq 0$, neither exponential receives a positive exponent from this branch extension. The exact BDF model retains Equation (35); the smoothing only supplies a globally finite NLP expression near the zero-settling threshold.

Several biochemical rates also contain a quotient defined by convention as zero when its denominator is zero, or a positive-part branch. Those conventions are continuous along many physical paths but are not differentiable expressions for IPOPT. For denominator type b , let $\mathcal{O}_b$ contain all applicable development-row and reactor-stage occurrences (only the row index for a plant-level denominator). Define

$$S_b = \max \left(1, \max_{(i,h) \in \mathcal{O}_b} |b_{ih}| \right) \quad \text{in the tabulated unit of } b, \quad \text{sdiv}_{\epsilon, S_b}(a, b) = \frac{ab}{b^2 + (\epsilon S_b)^2}. \quad (90)$$

For each original positive part $[v]_+$, define

$$S_{+,v} = \max \left(1, \max_{(i,h) \in \mathcal{O}_v} |v_{ih}| \right), \quad p_{\epsilon, S_{+,v}}(v) = \frac{1}{2} \left[v + \sqrt{v^2 + (\epsilon S_{+,v})^2} \right]. \quad (91)$$

All scales are frozen from the development targets. For each named denominator type, S_b pools absolute denominator values over every development row and all five reactor-stage occurrences where applicable; it is not stage specific. The persisted denominator scales, in order, are S_{NOx} for $S_{\text{NO2}} + S_{\text{NO3}}$, S_{FA} for $S_F + S_A$, S_{hyd} for $K_X X_H + X_S$, and

S_{PAO} for X_{PAO} . The scale $S_{+,PP}$ for the positive-part argument $K_{\max} - r_{PP}$ likewise pools that argument over every development row and reactor stage. In the combined NLP these expressions replace every zero-denominator share in Appendix A, including $\alpha_2, \alpha_3, \alpha_F, \alpha_A, \theta_X, r_{PP}, r_{PHA}$, and replace the positive part in $K_{\max} - r_{PP}$; all remaining occurrences follow the same stated rule.

The Clarifier reconstruction uses a separate extension because every feasible NLP point satisfies $s_F \geq S_0$, where $S_0 = 1 \text{ g TSS m}^{-3}$. With $\bar{s} = s/S_0$, define

$$\text{rec}_{S_0}(s) = \frac{1}{S_0} \begin{cases} \bar{s}^2 - 3\bar{s} + 3, & 0 \leq \bar{s} < 1, \\ 1/\bar{s}, & \bar{s} \geq 1. \end{cases} \quad (92)$$

This positive extension is C^2 , finite throughout the bound-respecting trial domain, and exactly $1/s$ throughout the feasible domain. The particulate Clarifier reconstruction is

$$c_{\text{cl},\ell,j}^{\text{sm}} = s_\ell c_{5,j} \text{rec}_{S_0}(s_F), \quad j \in \mathcal{X}, \quad (93)$$

with the same expression at the two outlet layers. Therefore $t_X^\top c_{\text{cl},\ell}^{\text{sm}} = s_\ell$ exactly at every feasible NLP point. The positive-part error is at most $\epsilon S_{+,v}/2$. The regularized biochemical quotient approaches a/b away from $b = 0$ but cannot uniformly reproduce the original zero-denominator convention near zero, and the storage-ratio derivatives can remain poorly conditioned when X_{PAO} is tiny. The preflight therefore records derivative finiteness, IPOPT restoration or factorization failures, and available Jacobian/Hessian condition estimates; it never clips a small denominator. The exact BDF replay uses the original biochemical definitions and explicitly tests this smoothing limitation.

The receiver-capacity switch at X_t is discontinuous and therefore needs a separate fixed transition. Put $\tau_X = \max(1 \text{ g m}^{-3}, 10^{-4} X_t) = 1 \text{ g m}^{-3}$ and

$$w_X(s) = \begin{cases} 0, & s \leq X_t - \tau_X, \\ 6\xi_X^5 - 15\xi_X^4 + 10\xi_X^3, & X_t - \tau_X < s < X_t + \tau_X, \quad \xi_X = \frac{s - (X_t - \tau_X)}{2\tau_X}, \\ 1, & s \geq X_t + \tau_X, \end{cases} \quad (94)$$

The smoothed downward limited flux is

$$\mathcal{J}_{\text{lim}}^{\text{sm}}(s_{\text{up}}, s_{\text{down}}) = [1 - w_X(s_{\text{down}})]j_s^{\text{sm}}(s_{\text{up}}) + w_X(s_{\text{down}}) \text{smin}_{\epsilon, S_J}[j_s^{\text{sm}}(s_{\text{up}}), j_s^{\text{sm}}(s_{\text{down}})]. \quad (95)$$

Substitution into the same advective flux construction gives every internal flux used by the NLP:

$$\mathcal{J}_{\ell+1/2}^{\text{sm}} = \begin{cases} -v_E s_{\ell+1} + \mathcal{J}_{\text{lim}}^{\text{sm}}(s_\ell, s_{\ell+1}), & \ell < \ell_F, \\ v_U s_\ell + \mathcal{J}_{\text{lim}}^{\text{sm}}(s_\ell, s_{\ell+1}), & \ell \geq \ell_F. \end{cases} \quad (96)$$

The physical boundary fluxes contain no settling branch and remain

$$\mathcal{J}_{1/2}^{\text{sm}} = -v_E s_1, \quad \mathcal{J}_{N_\ell+1/2}^{\text{sm}} = v_U s_{N_\ell}. \quad (97)$$

The quintic makes the transition C^2 . Outside the transition band, the stated bounds continue to apply to each individual smooth-minimum or smooth-maximum primitive; nested velocity and flux expressions accumulate those primitive errors and are not assigned an unproved single $\epsilon S/2$ bound. No uniform approximation bound to the original discontinuous switch exists inside the band, so the final exact branch replay is an essential validation rather than a numerical formality.

The combined-NLP residual scales are fixed before optimization. Write each raw layer row in concentration-rate form,

$$r_\ell^{\text{layer,sm}} = \frac{A_{\text{cl}}}{V_{\text{cl},\ell}} (\mathcal{J}_{\ell-1/2}^{\text{sm}} - \mathcal{J}_{\ell+1/2}^{\text{sm}}) + \mathbf{1}_{\{\ell=\ell_F\}} \frac{Q_0 q_C}{V_{\text{cl},\ell}} s_F. \quad (98)$$

For raw smooth balance row k , let $\mathcal{T}_{ik}^{(\text{dyn})}$ contain its signed physical terms at development state i : the hydraulic, individual stoichiometric-rate, and oxygen-transfer terms for each reactor component, or the two $A_{\text{cl}} \mathcal{J}^{\text{sm}}/V_{\text{cl},\ell}$ terms

and feed term for each layer. After topology and numerical coefficients are substituted, every algebraically nonzero term is included; a term that evaluates to zero at a particular development state remains in the list, whereas a structurally zero term is omitted. Thus all terms have units of concentration per day. Rows are ordered first by reactor 1–5 and component order, then by Clarifier layer 1–10. Define

$$(D_f)_{kk} = \max \left\{ 1 \text{ in row } k\text{'s tabulated concentration unit d}^{-1}, \left[\frac{1}{n_D} \sum_{i \in D} \frac{1}{|\mathcal{T}_{ik}^{(\text{dyn})}|} \sum_{t \in \mathcal{T}_{ik}^{(\text{dyn})}} t^2 \right]^{1/2} \right\}. \quad (99)$$

Thus D_f is positive, development-only, and independent of the start and trial state; the dynamic solver's start-dependent S_{dyn} is never reused for NLP or KKT scaling. Let $r_{\text{phys}}^{\text{sm}}(z, y; x)$ collect the 100 raw reactor balances and ten raw smoothed layer balances, and put $f_{\text{phys}}^{\text{sm}} = D_f^{-1} r_{\text{phys}}^{\text{sm}}$. These 110 rows force the controls and physical state to describe one smooth steady plant.

3.10. The Combined Physics-Constrained Statistical NLP

All required pieces can now be placed in one problem. For compactness, define the nine explicit inequality rows

$$g_{\text{NLP}}(z, \tilde{y}; x) = \begin{bmatrix} (1 - s_F)/(1 \text{ g TSS m}^{-3}) \\ (1 - B_X)/(1 \text{ g TSS m}^{-3}) \\ g_{\text{eng}}\{\vartheta(z), \chi(z, y; x)\} \\ d\{\vartheta(z), x, y\}/\delta - 1 \\ [\ell\{\vartheta(z), x\} - \ell_{\max}]/\max(\ell_{\max}, 10^{-12}) \end{bmatrix} \leq 0. \quad (100)$$

The first two rows keep the Clarifier reconstruction and solids-retention calculation in their physical domains; the next five impose process and equipment limits; the eighth keeps the reconstructed complete state inside the calibrated statistical envelope; and the ninth prevents extrapolation beyond the largest development leverage. With $y = \mu_y + S_y \tilde{y}$, the single optimization problem is

$$\begin{aligned} & \min_{z, \tilde{y}} \quad J\{\vartheta(z), \chi[\vartheta(z), y; x]\} \\ & \text{subject to} \quad 0 \leq z \leq 1, \quad y \geq 0, \\ & \quad f_{\text{phys}}^{\text{sm}}(z, y; x) = 0, \\ & \quad g_{\text{NLP}}(z, \tilde{y}; x) \leq 0. \end{aligned} \quad (101)$$

The objective in Equation (101) is exactly the engineering score J ; neither fidelity nor leverage is scalarized into it. The problem has 115 variables, 110 scaled smooth mechanistic equalities, and nine explicit inequalities in addition to variable bounds. Consequently, every scored response is reconstructed from a mechanistic state and remains supported by the calibrated statistical model.

3.11. Starts, Solver, KKT Replay, and Selection

Each case has exactly nine ordered, independent starts. Start 0 is (0.5, 0.5, 0.5, 0.5, 0.5). Starts 1–8 are rows 0–7, in generated order, of the exact eight-point, five-dimensional strength-1 Latin hypercube produced by the SplitMix64 and unbiased Fisher–Yates rules in the Mechanistic Design subsection, restarting at $q_0 = 271828$ and using coordinate order (H, a, r_I, r_R, w) . The same nine z values are persisted and replayed for every case.

For each z seed and case influent, the resulting 25-dimensional (ϑ, x) vector is standardized using the frozen development means and scales. The initial y is taken from the nearest accepted development row in Euclidean distance; the smallest development-row index breaks an exact tie. For initialization only, coordinate j is floored at $10^{-8} \max(1, s_{y,j})$ in its tabulated unit so every ratio is defined. This floor never changes stored data, equations, constraints, objective values, iterates, or acceptance replays. No primal or dual warm start is transferred between starts or cases.

All objective and constraint expressions are assembled symbolically in CasADi, which supplies exact first and second derivatives to IPOPT (Andersson et al., 2019; Wächter and Biegler, 2006). IPOPT uses MUMPS (Amestoy

et al., 2001), adaptive barrier updates, tol and constr_viol_tol of 10^{-8} , dual_inf_tol and compl_inf_tol of 10^{-6} , $\text{bound_relax_factor}=0$, and at most 5000 iterations. The internal vector $v = (z, \tilde{y})$ has true bounds $0 \leq z \leq 1$ and $\tilde{y} \geq -S_y^{-1}\mu_y$, with no finite state upper bound. Thus every constraint evaluation remains in the non-negative physical-state domain.

Solver status alone does not establish acceptance, so every returned point receives an independent KKT replay. Let v^L and v^U be the internal lower and upper bounds, omitting infinite entries from upper-bound norms, and let λ_x be CasADi's IPOPT bound multiplier. For two-sided z coordinates set $\mu_L = [-\lambda_x]_+$ and $\mu_U = [\lambda_x]_+$; for lower-only state coordinates set $\mu_L = -\lambda_x$ and $\mu_U = 0$. Let λ be the equality multiplier, μ_g the upper-bounded inequality multiplier, $h_{\text{NLP}} = f_{\text{phys}}^{\text{sm}}$, and $\sigma_g = -g_{\text{NLP}}$. Calculate

$$r_{\text{bnd}} = \max\{\| [v^L - v]_+ \|_\infty, \| [v - v^U]_+ \|_\infty, v^U < \infty\}, \quad (102)$$

$$r_{\text{pri}} = \max\{\| h_{\text{NLP}} \|_\infty, \| [g_{\text{NLP}}]_+ \|_\infty, r_{\text{bnd}}\}, \quad (103)$$

$$r_{\text{stat}} = \frac{\| \nabla J + (\nabla h_{\text{NLP}})^\top \lambda + (\nabla g_{\text{NLP}})^\top \mu_g - \mu_L + \mu_U \|_\infty}{1 + \| \nabla J \|_\infty + \| (\nabla h_{\text{NLP}})^\top \lambda \|_\infty + \| (\nabla g_{\text{NLP}})^\top \mu_g \|_\infty + \| \mu_L \|_\infty + \| \mu_U \|_\infty}, \quad (104)$$

$$r_{\text{dual}} = \max\left\{ \frac{\| [-\mu_g]_+ \|_\infty}{1 + \| \mu_g \|_\infty}, \frac{\| [-\mu_L]_+ \|_\infty}{1 + \| \mu_L \|_\infty} \right\}, \quad (105)$$

$$r_{\text{comp}} = \frac{\max\{\| \mu_g \odot \sigma_g \|_\infty, \| \mu_L \odot (v - v^L) \|_\infty, \| \mu_U \odot (v^U - v) \|_\infty, v^U < \infty\}}{1 + \max\{\| \mu_g \|_\infty \| \sigma_g \|_\infty, \| \mu_L \|_\infty \| v - v^L \|_\infty, \| \mu_U \|_\infty \| v^U - v \|_\infty, v^U < \infty\}}. \quad (106)$$

A start is accepted only when CasADi reports IPOPT `return_status` equal to `Solve_Succeeded` or `Solved_To_Acceptable_Level`, all unrounded primal and dual values are finite, $r_{\text{pri}} \leq 10^{-8}$, $\max_j [-y_j / \max(1, s_{y,j})]_+ \leq 10^{-10}$, and $r_{\text{stat}}, r_{\text{dual}}, r_{\text{comp}} \leq 10^{-6}$. Every other status and every solver exception fails, and no failed result is repaired. Among accepted starts, the smallest engineering objective is selected. Objectives within $10^{-10}(1 + |J_{\min}|)$ tie; ties use lexicographically smallest $(z_1, \dots, z_5)$ and then smallest start index. This one selected point is the best accepted multistart local candidate, not a global solution.

3.12. Independent Exact Replay and Decision Rule

Whenever the exact replay succeeds at controls ϑ , let $\chi_{\text{BDF}}(\vartheta, x)$ denote its reconstructed complete state and define

$$J_{\text{exact}}(\vartheta) = J[\vartheta, \chi_{\text{BDF}}(\vartheta, x)] \quad (107)$$

using the applicable case's fixed influent x , frozen objective weights, and development-only scales. The case-specific x is suppressed in the left-hand notation, and the quantity is undefined when the exact replay fails.

The selected combined-NLP result is tested at its controls by one independent execution of the exact nonsmoothed BDF generator. The replay always uses the prescribed fixed Start-1/conditional-Start-2 route from data generation; the NLP state is never used as the dynamic initial condition. The replay must meet the original residual and stability tests, both domain guards, and all five case-specific engineering constraints. In addition, the complete exact state must remain within the calibrated fidelity threshold:

$$d_{\text{BDF}} = \frac{1}{L} \left\| D_\chi^{-1} [\chi_{\text{BDF}} - \hat{\chi}(\vartheta_{\text{NLP}}, x)] \right\|_2^2 \leq \delta. \quad (108)$$

Failure of Equation (108) is classified separately as a statistical-fidelity-validation failure; it is an observed check at the selected control, not a conditional conformal guarantee. Residual/stability and domain/engineering failures are likewise distinguished.

For every successful exact solve, also report the selected-state discrepancies

$$\begin{aligned} E_{\text{state},2} &= \left[\frac{1}{L} \left\| D_\chi^{-1} (\chi_{\text{BDF}} - \chi_{\text{NLP}}) \right\|_2^2 \right]^{1/2}, \\ E_{\text{state},\infty} &= \left\| D_\chi^{-1} (\chi_{\text{BDF}} - \chi_{\text{NLP}}) \right\|_\infty, \\ \Delta J_{\text{state}} &= J(\vartheta_{\text{NLP}}, \chi_{\text{NLP}}) - J_{\text{exact}}(\vartheta_{\text{NLP}}). \end{aligned} \quad (109)$$

These three discrepancies are descriptive and add no further acceptance threshold.

The exact physical state y_{BDF} must also agree with the selected smooth state y_{NLP} under the same frozen S_y :

$$\left\| S_y^{-1}(y_{\text{BDF}} - y_{\text{NLP}}) \right\|_{\infty} \leq 10^{-3}. \quad (110)$$

Failure of Equation (110) is a branch/smoothing-validation failure. The selected point is validated only when the single exact replay passes residual, stability, domain, engineering, fidelity, and branch-agreement checks. If any check fails, the case is reported with that failure and no alternative start is substituted after selection. The combined-NLP point and its exact-validation outcome are therefore the single reported decision route.

3.13. Robustness, Reporting, and Reproducibility

After the nominal problem and all scales are frozen, 100 fresh-influent profiles are generated by the Latin-hypercube construction in the Mechanistic Design subsection, restricted to the 20 influent dimensions and restarting its 64-bit state at $q_0 = 314159$. The `test_2000` and unit profiles use robustness seeds 2000314159 and 600314159, respectively; the nine-start seed 271828 remains common because it defines the optimization algorithm rather than data. No response from these profiles enters estimation, scaling, assessment, objective definition, fidelity or leverage limits, or numerical tolerances. For every case, the archive retains all nine start records and KKT replays, the selected controls and state, six unweighted and weighted objective components, all nine explicit inequality rows, the one exact-BDF outcome, Clarifier and reactor diagnostics, evaluation counts, durations, and peak memory.

Failures are not removed from summaries. One outcome table classifies the combined route using the first applicable row in this order: no accepted NLP start, exact integration failure, residual/stability failure, domain/engineering failure, statistical-fidelity failure, branch/smoothing disagreement, and validated candidate. These categories are mutually exclusive, the robustness table uses all 100 cases as its denominator, and an unobserved category remains as an explicit zero-count row.

For any scalar with n defined values $r_{[1]} \leq \dots \leq r_{[n]}$, set $Q_p = r_{[pn]}$. Each continuous-summary row states its eligible n and reports the mean, $Q_{0.25}$, $Q_{0.50}$, $Q_{0.75}$, interquartile range, $Q_{0.95}$, and maximum. The rows include the five controls; all six unweighted and weighted objective components; the combined-NLP and exact engineering objectives; d/δ , d_{BDF}/δ , leverage, and all three discrepancies in Equation (109); both domain guards; all five engineering quantities; and overflow COD, TN, TP, and TSS. Clarifier rows include component and composite outlet errors, particulate recoveries, layer profile and inventory, SRT, SOR, SLR, underflow TSS, and mass closure. Reactor rows include the stagewise dissolved-oxygen, nitrogen, phosphorus, and solids profiles and the largest exact kinetic-balance residual.

The activity table uses unrounded selected NLP results. A lower or upper control bound is active when $z_j \leq 10^{-6}$ or $1 - z_j \leq 10^{-6}$, respectively. A scaled inequality is active when its value lies in $[-10^{-3}, 10^{-8}]$. Eligible counts and fractions are reported for both bounds of every control and for each of the nine explicit rows: two domain guards, five engineering rows, fidelity, and leverage. Exact-BDF engineering activity is reported separately and never substituted for NLP activity.

The underflow-TSS sensitivity table has three rows for 12,000, 15,000, and 20,000 g TSS m^{-3} ; the 15,000 row is the nominal result. The objective-weight table has the ten cases in Equation (81). Both tables show all six coefficients and unweighted components, the selected controls, the single exact-replay outcome, all five exact engineering quantities, fidelity, leverage, branch agreement, and the validation classification. The TSS table applies its row-specific underflow limit, whereas every wasted-solids objective denominator remains 15,000 g TSS m^{-3} .

Timing uses a monotonic nanosecond-resolution counter and one numerical thread per NLP start. Processor, exact installed memory in bytes, operating system, interpreter, CasADi, IPOPT, MUMPS, linear-algebra and integration-library versions, and the data-generation worker count are recorded. The three mechanistic blocks, 14,000-row fit, calibration, assessment, each NLP start, and each exact replay are timed during their scientific execution and are not repeated merely for timing. Tables report count, total, mean, median, interquartile range, nearest-rank 95th percentile, maximum, and maximum resident memory by stage; they also show the empirical projection, safety factor, planned maximum workload, and realized workload. The 256-row BDF preflight and 23-case NLP panel remain in their scientific stages and their cached results are reused.

The permanent study record contains generator states and draw counts, block membership, component and feature orders, topology, all parameter values, the nine ordered starts, numerical options and tolerances, fitted coefficients, frozen scales, calibration order statistic, assessment values, the complete start archive, selected candidate, exact replay,

and table-ready diagnostics. A terminal replay reloads the stored unrounded exact state and independently recomputes the mechanistic residual, stability result, two domain guards, five engineering rows, exact fidelity, and branch agreement before any “validated” classification is released. It also recomputes the stored NLP algebraic and KKT checks. Only a complete record satisfying these contracts is eligible for scientific reporting.

4. Results and Discussion

Several structural results follow before numerical values are considered. The flow definitions close at the plant boundary, so MLR and RAS change internal concentrations without appearing as generation or loss. Every accepted local point satisfies all 110 scaled smooth reactor-and-Clarifier balances, physical bounds, two domain guards, five engineering rows, calibrated fidelity, and development support. Ten layer concentrations make Clarifier inventory and whole-plant SRT traceable rather than inferred from an outlet alone.

The statistical evidence must be read in its declared order. Development data determine the single 351-feature, 170-response estimator and every scale. Calibration determines one squared standardized-RMS fidelity threshold. Assessment then reports raw errors and the empirical fidelity-coverage gate without changing anything upstream. Split conformal supports a marginal statement for a future iid-uniform design row conditional on development; it does not certify the adaptively optimized state. The optimized point is instead tested by its independent exact BDF replay.

The result is one joint statistical-and-physical local solution, not a corrected free prediction. The complete response is reconstructed from the mechanistic state, while hard fidelity and leverage constraints connect it to the development evidence. Because J is the sole objective, statistical proximity cannot be traded against engineering performance after its admissible threshold has been met. The independent nonsmoothed replay adjudicates residual, stability, domain, engineering feasibility, exact statistical fidelity, biochemical regularization, and the discontinuous receiver branch at the selected controls.

Changing H at fixed Q_0 changes installed reactor volume, so the problem is joint design and operation; the number of stages remains prescribed. Nine starts reduce dependence on a single initialization but do not establish global optimality, and IPOPT’s accepted KKT point need not be unique. The selected point is therefore described only as the best accepted candidate among the declared starts, subject to its one exact validation outcome.

The two sensitivity groups answer process-design questions rather than provide additional model fitting. The three underflow-TSS limits expose the consequence of Clarifier underflow capacity. The ten one-at-a-time weight cases show how controls and unweighted performance components respond to stated engineering preferences. Robustness profiles show the distribution of local decisions and validation outcomes across influent variation; the separate failure tables prevent a favorable continuous summary from concealing unsolved or invalid cases.

5. Conclusion

The recycling mixer–reactor–Clarifier system is optimized as one simultaneous 110-state and five-control decision. A fixed development-only statistical surrogate supplies the complete-response center; independent calibration supplies a conformal fidelity threshold; untouched assessment gates raw predictive use; and no subsequent refit or model selection changes that information boundary.

Smooth ASM2d-TSN reactor and Clarifier balances, physical bounds, engineering restrictions, calibrated fidelity, and development support define the combined NLP; the engineering criterion alone is minimized. One exact fixed-start BDF replay decides whether its selected point belongs to the intended stable nonsmoothed branch and satisfies domain, engineering, and fidelity requirements. Nine deterministic starts support reproducibility but not a global claim. The single selected candidate, its validation classification, sensitivity cases, and robustness outcomes therefore have explicit and appropriately limited meanings.

A. Complete ASM2d-TSN Mechanistic Definition

A.1. Rate functions and process rates

For a non-negative concentration z and a positive half-saturation constant K , define

$$M(z; K) = \frac{z}{K + z}, \quad \overline{M}(z; K) = \frac{K}{K + z}. \quad (111)$$

Let $S_{\text{NOx}} = S_{\text{NO2}} + S_{\text{NO3}}$ and define the electron-acceptor and carbon shares

$$\alpha_2 = \frac{S_{\text{NO2}}}{S_{\text{NOx}}}, \quad \alpha_3 = \frac{S_{\text{NO3}}}{S_{\text{NOx}}}, \quad \alpha_F = \frac{S_F}{S_F + S_A}, \quad \alpha_A = \frac{S_A}{S_F + S_A}. \quad (112)$$

A share is zero when its denominator is zero. Hydrolysable substrate availability and the PAO storage ratios are

$$\theta_X = \frac{X_S}{K_X X_H + X_S}, \quad r_{PP} = \frac{X_{PP}}{X_{PAO}}, \quad r_{PHA} = \frac{X_{PHA}}{X_{PAO}}, \quad (113)$$

The value of θ_X is zero when both X_H and X_S are zero. When $X_{PAO} = 0$, both storage ratios are defined as zero; this convention makes every PAO rate evaluable at the non-negative boundary.

$$\pi_{PP} = M(r_{PP}; K_{PP}), \quad \pi_{PHA} = M(r_{PHA}; K_{PHA}), \quad c_{PP} = \frac{[K_{\max} - r_{PP}]_+}{K_{IPP} + [K_{\max} - r_{PP}]_+}. \quad (114)$$

All PAO rates that multiply X_{PAO} are defined as zero when $X_{PAO} = 0$. The nutrient factors used repeatedly below are

$$L_H = M(S_{\text{NH4}}; K_{\text{NH4},H})M(S_{\text{PO4}}; K_{\text{PO4},H})M(S_{\text{ALK}}; K_{\text{ALK},H}), \quad (115)$$

$$L_P = M(S_{\text{NH4}}; K_{\text{NH4},PAO})M(S_{\text{PO4}}; K_{\text{PO4},PAO})M(S_{\text{ALK}}; K_{\text{ALK},PAO}), \quad (116)$$

$$L_N = M(S_{\text{PO4}}; K_{\text{PO4},\text{nit}})M(S_{\text{ALK}}; K_{\text{ALK},\text{nit}}). \quad (117)$$

The hydrolysis, heterotrophic growth, fermentation, and heterotroph-decay rates are

$$\rho_1 = K_H M(S_O; K_{O,\text{hyd}}) \theta_X X_H, \quad (118)$$

$$\rho_2 = \eta_{\text{hyd},\text{NO2}} K_H \overline{M}(S_O; K_{O,\text{hyd}}) M(S_{\text{NO2}}; K_{\text{NO2},\text{hyd}}) \alpha_2 \theta_X X_H, \quad (119)$$

$$\rho_3 = \eta_{\text{hyd},\text{NO3}} K_H \overline{M}(S_O; K_{O,\text{hyd}}) M(S_{\text{NO3}}; K_{\text{NO3},\text{hyd}}) \alpha_3 \theta_X X_H, \quad (120)$$

$$\rho_4 = \eta_{\text{hyd},f_e} K_H \overline{M}(S_O; K_{O,\text{hyd}}) \overline{M}(S_{\text{NOx}}; K_{\text{NOx},\text{hyd}}) \theta_X X_H, \quad (121)$$

$$\rho_5 = \mu_H M(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H X_H, \quad (122)$$

$$\rho_6 = \mu_H M(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H X_H, \quad (123)$$

$$\rho_7 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H \eta_{H,\text{NO3}} M(S_{\text{NO3}}; K_{\text{NO3},H}) \alpha_3 X_H, \quad (124)$$

$$\rho_8 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H \eta_{H,\text{NO2}} M(S_{\text{NO2}}; K_{\text{NO2},H}) \alpha_2 X_H, \quad (125)$$

$$\rho_9 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H \eta_{H,\text{NO3}} M(S_{\text{NO3}}; K_{\text{NO3},H}) \alpha_3 X_H, \quad (126)$$

$$\rho_{10} = \mu_H \overline{M}(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H \eta_{H,\text{NO2}} M(S_{\text{NO2}}; K_{\text{NO2},H}) \alpha_2 X_H, \quad (127)$$

$$\rho_{11} = q_{f_e} \overline{M}(S_O; K_{O,H}) \overline{M}(S_{\text{NOx}}; K_{\text{NOx},H}) M(S_F; K_{f_e}) M(S_{\text{ALK}}; K_{\text{ALK},H}) X_H, \quad (128)$$

$$\rho_{12} = b_H X_H. \quad (129)$$

Here q_{f_e} enters ρ_{11} directly as a rate constant with units d^{-1} .

The PAO storage, growth, and decay rates are

$$\rho_{13} = q_{PHA} M(S_A; K_A) \overline{M}(S_O; K_{O,PAO}) \overline{M}(S_{\text{NOx}}; K_{\text{NOx},PAO}) M(S_{\text{ALK}}; K_{\text{ALK},PAO}) \pi_{PP} X_{PAO}, \quad (130)$$

$$\rho_{14} = q_{PP} M(S_O; K_{O,PAO}) M(S_{\text{PO4}}; K_{PS}) M(S_{\text{ALK}}; K_{\text{ALK},PAO}) \pi_{PHA} c_{PP} X_{PAO}, \quad (131)$$

$$\rho_{15} = q_{PP} \overline{M}(S_O; K_{O,PAO}) M(S_{\text{PO4}}; K_{PS}) M(S_{\text{ALK}}; K_{\text{ALK},PAO}) \pi_{PHA} c_{PP} \quad (132)$$

$$\times \eta_{PAO,\text{NO3}} M(S_{\text{NO3}}; K_{\text{NO3},PAO}) \alpha_3 X_{PAO}, \quad (133)$$

$$\rho_{16} = q_{PP} \overline{M}(S_O; K_{O,PAO}) M(S_{\text{PO4}}; K_{PS}) M(S_{\text{ALK}}; K_{\text{ALK},PAO}) \pi_{PHA} c_{PP} \quad (134)$$

$$\times \eta_{PAO,\text{NO2}} M(S_{\text{NO2}}; K_{\text{NO2},PAO}) \alpha_2 X_{PAO}, \quad (135)$$

$$\rho_{17} = \mu_{PAO} M(S_O; K_{O,PAO}) L_P \pi_{PHA} X_{PAO}, \quad (136)$$

$$\rho_{18} = \mu_{PAO} \overline{M}(S_O; K_{O,PAO}) L_P \pi_{PHA} \eta_{PAO,\text{NO3}} M(S_{\text{NO3}}; K_{\text{NO3},PAO}) \alpha_3 X_{PAO}, \quad (137)$$

$$\rho_{19} = \mu_{PAO} \overline{M}(S_O; K_{O,PAO}) L_P \pi_{PHA} \eta_{PAO,NO_2} M(S_{NO_2}; K_{NO_2,PAO}) \alpha_2 X_{PAO}, \quad (138)$$

$$\rho_{20} = b_{PAO} M(S_{ALK}; K_{ALK,PAO}) X_{PAO}, \quad (139)$$

$$\rho_{21} = b_{PP} M(S_{ALK}; K_{ALK,PAO}) X_{PP}, \quad (140)$$

$$\rho_{22} = b_{PHA} M(S_{ALK}; K_{ALK,PAO}) X_{PHA}. \quad (141)$$

The oxygen and NOx inhibition in ρ_{13} makes PHA storage anaerobic. The nitrifier and chemical rates are

$$\rho_{23} = \mu_{AOB} M(S_O; K_{O,AOB}) M(S_{NH_4}; K_{NH_4,AOB}) L_N X_{AOB}, \quad (142)$$

$$\rho_{24} = \mu_{NOB} M(S_O; K_{O,NOB}) M(S_{NO_2}; K_{NO_2,NOB}) L_N X_{NOB}, \quad (143)$$

$$\rho_{25} = b_{AOB} X_{AOB}, \quad \rho_{26} = b_{NOB} X_{NOB}, \quad (144)$$

$$\rho_{27} = k_{PRE} S_{PO_4} X_{MeOH}, \quad (145)$$

$$\rho_{28} = k_{RED} i_{PMeP} M(S_{ALK}; K_{ALK,chem}) X_{MeP}. \quad (146)$$

The factor $i_{PMeP} = 1/4.87$ g P g TSS⁻¹ converts redissolving X_{MeP} to the phosphorus basis used by process 28. At a component state c , the process-rate vector used in the reactor equations is

$$\varrho(c) = [\rho_1(c), \dots, \rho_{28}(c)]^\top \in \mathbb{R}_+^{28}, \quad \varrho_i = \varrho(c_i). \quad (147)$$

Rates $\rho_{1:13}$, $\rho_{17:20}$, $\rho_{22:26}$ have a COD mass basis; $\rho_{14:16}$, ρ_{21} , and $\rho_{27:28}$ have a phosphorus mass basis. Each row of ν below converts its rate basis into the units of all 20 component coordinates, so $\nu^\top \varrho(c)$ is a component-rate vector. Oxygen transfer in reactor i is

$$\omega_i = k_L a_i (S_{O,sat} - c_{i,S_O}), \quad S_{O,sat} = 8.5 \text{ g O}_2 \text{ m}^{-3}, \quad (148)$$

with $k_L a_i = 0 \text{ d}^{-1}$ for $i = 1, 2$ and $k_L a_i = 47a \text{ d}^{-1}$ for $i = 3, 4, 5$.

A.2. Stoichiometric matrix and invariant operator

Let $e_j \in \mathbb{R}^F$ be the column coordinate vector for component j ; its transpose is the corresponding row selector. Omitted coefficients below are zero. Define the preliminary process rows

$$\bar{\nu}_{1:4} = (1 - f_{SI}) e_{S_F}^\top + f_{SI} e_{S_I}^\top - e_{X_S}^\top, \quad (149)$$

$$\bar{\nu}_5 = -Y_H^{-1} e_{S_F}^\top + (1 - Y_H^{-1}) e_{S_O}^\top + e_{X_H}^\top, \quad (150)$$

$$\bar{\nu}_6 = -Y_H^{-1} e_{S_A}^\top + (1 - Y_H^{-1}) e_{S_O}^\top + e_{X_H}^\top, \quad (151)$$

$$\bar{\nu}_7 = -Y_H^{-1} e_{S_F}^\top + \beta_H (e_{S_{NO_2}} - e_{S_{NO_3}})^\top + e_{X_H}^\top, \quad (152)$$

$$\bar{\nu}_8 = -Y_H^{-1} e_{S_F}^\top + \gamma_H (e_{S_{N_2}} - e_{S_{NO_2}})^\top + e_{X_H}^\top, \quad (153)$$

$$\bar{\nu}_9 = -Y_H^{-1} e_{S_A}^\top + \beta_H (e_{S_{NO_2}} - e_{S_{NO_3}})^\top + e_{X_H}^\top, \quad (154)$$

$$\bar{\nu}_{10} = -Y_H^{-1} e_{S_A}^\top + \gamma_H (e_{S_{N_2}} - e_{S_{NO_2}})^\top + e_{X_H}^\top, \quad (155)$$

$$\bar{\nu}_{11} = e_{S_A}^\top - e_{S_F}^\top, \quad (156)$$

$$\bar{\nu}_{12} = f_{XI} e_{X_I}^\top + (1 - f_{XI}) e_{X_S}^\top - e_{X_H}^\top, \quad (157)$$

$$\bar{\nu}_{13} = -e_{S_A}^\top - Y_{PO_4} e_{X_{PP}}^\top + e_{X_{PHA}}^\top, \quad (158)$$

$$\bar{\nu}_{14} = -Y_{PHA} e_{S_O}^\top + e_{X_{PP}}^\top - Y_{PHA} e_{X_{PHA}}^\top, \quad (159)$$

$$\bar{\nu}_{15} = \beta_S (e_{S_{NO_2}} - e_{S_{NO_3}})^\top + e_{X_{PP}}^\top - Y_{PHA} e_{X_{PHA}}^\top, \quad (160)$$

$$\bar{\nu}_{16} = \gamma_S (e_{S_{N_2}} - e_{S_{NO_2}})^\top + e_{X_{PP}}^\top - Y_{PHA} e_{X_{PHA}}^\top, \quad (161)$$

$$\bar{\nu}_{17} = (1 - Y_{PAO}^{-1}) e_{S_O}^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1} e_{X_{PHA}}^\top, \quad (162)$$

$$\bar{\nu}_{18} = \beta_P (e_{S_{NO_2}} - e_{S_{NO_3}})^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1} e_{X_{PHA}}^\top, \quad (163)$$

$$\bar{v}_{19} = \gamma_P(e_{S_{N2}} - e_{S_{NO2}})^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1}e_{X_{PHA}}^\top, \quad (164)$$

$$\bar{v}_{20} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{PAO}}^\top, \quad (165)$$

$$\bar{v}_{21} = -e_{X_{PP}}^\top, \quad \bar{v}_{22} = e_{S_A}^\top - e_{X_{PHA}}^\top, \quad (166)$$

$$\bar{v}_{23} = -\frac{3.43 - Y_{AOB}}{Y_{AOB}}e_{S_O}^\top + Y_{AOB}^{-1}e_{S_{NO2}}^\top + e_{X_{AOB}}^\top, \quad (167)$$

$$\bar{v}_{24} = -\frac{1.14 - Y_{NOB}}{Y_{NOB}}e_{S_O}^\top - Y_{NOB}^{-1}e_{S_{NO2}}^\top + Y_{NOB}^{-1}e_{S_{NO3}}^\top + e_{X_{NOB}}^\top, \quad (168)$$

$$\bar{v}_{25} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{AOB}}^\top, \quad (169)$$

$$\bar{v}_{26} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{NOB}}^\top, \quad (170)$$

$$\bar{v}_{27} = -e_{S_{PO4}}^\top - \gamma_{OH}e_{X_{MeOH}}^\top + i_{PMeP}^{-1}e_{X_{MeP}}^\top, \quad (171)$$

$$\bar{v}_{28} = e_{S_{PO4}}^\top + \gamma_{OH}e_{X_{MeOH}}^\top - i_{PMeP}^{-1}e_{X_{MeP}}^\top, \quad (172)$$

where

$$\beta_H = \frac{1 - Y_H}{(8/7)Y_H}, \quad \gamma_H = \frac{1 - Y_H}{1.72Y_H}, \quad \beta_S = \frac{Y_{PHA}}{8/7}, \quad \gamma_S = \frac{Y_{PHA}}{1.72}, \quad (173)$$

$$\beta_P = \frac{1 - Y_{PAO}}{(8/7)Y_{PAO}}, \quad \gamma_P = \frac{1 - Y_{PAO}}{1.72Y_{PAO}}. \quad (174)$$

Define

$$\mathcal{N} = \{S_F, S_I, S_{N2}, S_{NO2}, S_{NO3}, X_I, X_S, X_H, X_{PAO}, X_{AOB}, X_{NOB}\}, \quad (175)$$

$$\mathcal{P} = \{S_F, S_I, X_I, X_S, X_H, X_{PAO}, X_{AOB}, X_{NOB}, X_{PP}, X_{MeP}\}. \quad (176)$$

Nitrogen continuity completes every S_{NH4} coefficient:

$$v_{p,S_{NH4}} = - \sum_{j \in \mathcal{N}} n_j \bar{v}_{p,j}, \quad (177)$$

where $n_{S_F} = i_{NSF}$, $n_{S_I} = i_{NSI}$, $n_{S_{N2}} = n_{S_{NO2}} = n_{S_{NO3}} = 1$, $n_{X_I} = i_{NXI}$, $n_{X_S} = i_{NXS}$, and $n_{X_H} = n_{X_{PAO}} = n_{X_{AOB}} = n_{X_{NOB}} = i_{NBM}$. Except for the explicitly specified rows 27–28, phosphorus continuity gives

$$v_{p,S_{PO4}} = - \sum_{j \in \mathcal{P}} p_j \bar{v}_{p,j}, \quad (178)$$

where $p_{S_F} = i_{PSF}$, $p_{S_I} = i_{PSI}$, $p_{X_I} = i_{PXI}$, $p_{X_S} = i_{PXS}$, $p_{X_H} = p_{X_{PAO}} = p_{X_{AOB}} = p_{X_{NOB}} = i_{PBM}$, $p_{X_{PP}} = 1$, and $p_{X_{MeP}} = i_{PMeP}$. All weights not listed are zero. Alkalinity is then

$$v_{p,S_{ALK}} = \frac{v_{p,S_{NH4}}}{14} - \frac{v_{p,S_{NO2}}}{14} - \frac{v_{p,S_{NO3}}}{14} + \frac{v_{p,S_{PO4}}}{31}. \quad (179)$$

For every other coordinate $j \notin \{S_{NH4}, S_{PO4}, S_{ALK}\}$, set $v_{p,j} = \bar{v}_{p,j}$; rows 27–28 retain their displayed S_{PO4} entries. Together with the preliminary rows, these rules define all entries of $v \in \mathbb{R}^{28 \times 20}$.

The deterministic physical conservation rows are

$$b_{SI} = e_{S_I}^\top, \quad (180)$$

$$b_N = e_{S_{NH4}}^\top + e_{S_{NO2}}^\top + e_{S_{NO3}}^\top + e_{S_{N2}}^\top + i_{NSI}e_{S_I}^\top + i_{NSF}e_{S_F}^\top + i_{NXI}e_{X_I}^\top + i_{NXS}e_{X_S}^\top \quad (181)$$

$$+ i_{NBM}(e_{X_H} + e_{X_{PAO}} + e_{X_{AOB}} + e_{X_{NOB}})^\top, \quad (182)$$

$$b_P = e_{S_{PO4}}^\top + i_{PSI}e_{S_I}^\top + i_{PSF}e_{S_F}^\top + i_{PXI}e_{X_I}^\top + i_{PXS}e_{X_S}^\top + e_{X_{PP}}^\top \quad (183)$$

$$+ i_{PBM}(e_{X_H} + e_{X_{PAO}} + e_{X_{AOB}} + e_{X_{NOB}})^\top + i_{PMeP}e_{X_{MeP}}^\top, \quad (184)$$

$$b_{alk} = e_{S_{ALK}}^\top - (1/14)e_{S_{NH4}}^\top + (1/14)e_{S_{NO2}}^\top + (1/14)e_{S_{NO3}}^\top - (1/31)e_{S_{PO4}}^\top, \quad (185)$$

$$b_{metal} = \gamma_{OH}e_{X_{MeP}}^\top + i_{PMeP}^{-1}e_{X_{MeOH}}^\top. \quad (186)$$

With

$$B = \begin{bmatrix} b_{SI} \\ b_N \\ b_P \\ b_{alk} \\ b_{metal} \end{bmatrix} \in \mathbb{R}^{5 \times 20} \quad (187)$$

and b_k denoting row k , define the deterministic row-normalized operator

$$D_A = \text{Diag}(\|b_1\|_2^{-1}, \dots, \|b_5\|_2^{-1}), \quad A = D_A B. \quad (188)$$

Thus $Av^\top = 0$, $Ae_{S_O} = 0$, $\text{rank}(A) = 5$, and every row of A has unit Euclidean norm. No linear combination is taken between the five named inventories.

A.3. Composite maps

In the component order of Equation (34), the rows of the composite map are COD, reported TN, TP, and TSS:

$$I_{\text{comp}} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0.01 & 0 & 0.02 & 0 & 0.07 & 0.07 & 0 & 0 & 0.07 & 0.07 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0.01 & 0 & 0.02 & 0.02 & 1 & 0 & 0.02 & 0.02 & i_{PMeP} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.75 & 0.75 & 0.90 & 0.90 & 3.23 & 0.60 & 0.90 & 0.90 & 1 \end{bmatrix}. \quad (189)$$

Reported TN excludes S_{N2} because it represents converted dinitrogen rather than a regulated aqueous nitrogen species. The TSS vector is the transpose of the last row:

$$t_X = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.75, 0.75, 0.90, 0.90, 3.23, 0.60, 0.90, 0.90, 1, 1)^\top. \quad (190)$$

A.4. Fixed biological and chemical parameters

Table 5: Complete ASM2d-TSN parameter set.

Group	Symbol	Value	Unit
Stoichiometry	Y_H	0.625	g COD g COD ⁻¹
	Y_{PAO}	0.625	g COD g COD ⁻¹
	Y_{PHA}	0.20	g COD g P ⁻¹
	Y_{PO4}	0.40	g P g COD ⁻¹
	Y_{AOB}	0.18	g COD g N ⁻¹
	Y_{NOB}	0.08	g COD g N ⁻¹
	f_{SI}	0	—
	f_{XI}	0.10	—
Composition	i_{NSI}, i_{NSF}	0.01, 0	g N g COD ⁻¹
	$i_{NXI}, i_{NXS}, i_{NBM}$	0.02, 0, 0.07	g N g COD ⁻¹
	i_{PSI}, i_{PSF}	0, 0	g P g COD ⁻¹
	$i_{PXI}, i_{PXS}, i_{PBM}$	0.01, 0, 0.02	g P g COD ⁻¹
	i_{PMeP}	1/4.87	g P g TSS ⁻¹
	$i_{TSS,XI}, i_{TSS,XS}$	0.75, 0.75	g TSS g COD ⁻¹
	$i_{TSS,BM}, i_{TSS,PHA}$	0.90, 0.60	g TSS g COD ⁻¹

Group	Symbol	Value	Unit
Hydrolysis	$i_{TSS,PP}$	3.23	g TSS g P ⁻¹
	K_H	3.0	d ⁻¹
	$\eta_{hyd,NO3}, \eta_{hyd,NO2}, \eta_{hyd,fe}$	0.6, 0.6, 0.4	–
	$K_{O,hyd}$	0.20	g O ₂ m ⁻³
	$K_{NO3,hyd}, K_{NO2,hyd}, K_{NOx,hyd}$	0.5, 0.5, 0.5	g N m ⁻³
Heterotrophs	K_X	0.10	g X_S g X_H^{-1}
	μ_H, q_{fe}, b_H	6.0, 3.0, 0.40	d ⁻¹
	$\eta_{H,NO3}, \eta_{H,NO2}$	0.9, 0.9	–
	$K_{O,H}$	0.20	g O ₂ m ⁻³
	K_F, K_{fe}, K_A	4, 4, 4	g COD m ⁻³
	$K_{NO3,H}, K_{NO2,H}, K_{NOx,H}$	0.5, 0.5, 0.5	g N m ⁻³
	$K_{NH4,H}$	0.05	g N m ⁻³
	$K_{PO4,H}$	0.01	g P m ⁻³
	$K_{ALK,H}$	0.10	mol HCO ₃ ⁻ m ⁻³
	q_{PHA}	5.0	d ⁻¹
PAO	q_{PP}	0.60	g P (g COD d) ⁻¹
	μ_{PAO}	0.56	d ⁻¹
	$\eta_{PAO,NO3}, \eta_{PAO,NO2}$	0.07, 0.90	–
	b_{PAO}, b_{PP}, b_{PHA}	0.20, 0.20, 0.20	d ⁻¹
	$K_{O,PAO}$	0.20	g O ₂ m ⁻³
	$K_{NO3,PAO}, K_{NO2,PAO}, K_{NOx,PAO}$	0.5, 0.5, 0.5	g N m ⁻³
	$K_{NH4,PAO}$	0.05	g N m ⁻³
	$K_{PS}, K_{PO4,PAO}$	0.20, 0.01	g P m ⁻³
	$K_{ALK,PAO}$	0.10	mol HCO ₃ ⁻ m ⁻³
	K_{PP}, K_{max}, K_{IPP}	0.01, 0.34, 0.02	g P g COD ⁻¹
	K_{PHA}	0.01	g COD g COD ⁻¹
	μ_{AOB}, μ_{NOB}	1.81, 1.52	d ⁻¹
	b_{AOB}, b_{NOB}	0.20, 0.17	d ⁻¹
	$K_{O,AOB}, K_{O,NOB}$	0.74, 1.75	g O ₂ m ⁻³
	$K_{NH4,AOB}, K_{NO2,NOB}$	0.5, 0.5	g N m ⁻³
Nitrifiers	$K_{ALK,nit}$	0.5	mol HCO ₃ ⁻ m ⁻³
	$K_{PO4,nit}$	0.01	g P m ⁻³
	k_{PRE}	1.0	m ³ (g TSS d) ⁻¹
	k_{RED}	0.60	d ⁻¹
	γ_{OH}	3.45	g TSS g P ⁻¹
Chemical	$K_{ALK,chem}$	0.50	mol HCO ₃ ⁻ m ⁻³

CRedit authorship contribution statement

The sole author, Egberto F. Selerio Jr., contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review & editing, visualization, supervision, project administration, and funding acquisition.

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Data availability

All equations, parameter values, coordinate orders, sampling rules, estimators, acceptance tests, and nonlinear-programming rules needed to reproduce the calculation are contained in this article. The generated 20,000-state table, fitted coefficient matrix, assessment table, and complete optimization archives will be deposited as a public research-data record when the numerical study is complete.

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